

3.21.39 SQL: GATE CSE 2017 | Set 2 | Question: 46



Consider the following database table named top_scorer.

top_scorer		
player	country	goals
Klose	Germany	16
Ronaldo	Brazil	15
G Muller	Germany	14
Fontaine	France	13
Pele	Brazil	12
Klinsmann	Germany	11
Kocsis	Hungary	11
Batistuta	Argentina	10
Cubillas	Peru	10
Lato	Poland	10
Lineker	England	10
T Muller	Germany	10
Rahn	Germany	10

Consider the following SQL query:

```
SELECT ta.player FROM top_scorer AS ta
WHERE ta.goals > ALL (SELECT tb.goals
FROM top_scorer AS tb
WHERE tb.country = 'Spain')
AND ta.goals > ANY (SELECT tc.goals
FROM top_scorer AS tc
WHERE tc.country != 'Germany')
```

The number of tuples returned by the above SQL query is _____.

3.21.40 SQL: GATE CSE 2018 | Question: 12



Consider the following two tables and four queries in SQL.

Book(isbn, bname), Stock(isbn, copies)

Query 1:

```
SELECT B.isbn, S.copies FROM Book B INNER JOIN Stock S ON B.isbn=S.isbn;
```

Query 2:

```
SELECT B.isbn, S.copies FROM Book B LEFT OUTER JOIN Stock S ON B.isbn=S.isbn;
```

Query 3:

```
SELECT B.isbn, S.copies FROM Book B RIGHT OUTER JOIN Stock S ON B.isbn=S.isbn
```

Query 4:

```
SELECT B.isbn, S.copies FROM Book B FULL OUTER JOIN Stock S ON B.isbn=S.isbn
```

Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?

A. Query 1

B. Query 2

C. Query 3

D. Query 4

gatecse-2018 databases sql easy one-mark

Answer key

3.21.41 SQL: GATE CSE 2019 Question: 51



A relational database contains two tables Student and Performance as shown below:

Table: student		Table: Performance		
Roll_no	Student_name	Roll_no	Subject_code	Marks
1	Amit	1	A	86
2	Priya	1	B	95
3	Vinit	1	C	90
4	Rohan	2	A	89
5	Smita	2	C	92
		3	C	80

The primary key of the Student table is Roll_no. For the performance table, the columns Roll_no and Subject_code together form the primary key. Consider the SQL query given below:

```
SELECT S.Student_name, sum(P.Marks)
FROM Student S, Performance P
WHERE P.Marks > 84
GROUP BY S.Student_name;
```

The number of rows returned by the above SQL query is _____

gatecse-2019 numerical-answers databases sql two-marks

Answer key

3.21.42 SQL: GATE CSE 2020 Question: 13



Consider a relational database containing the following schemas.

Catalogue			Suppliers			Parts		
sno	pno	cost	sno	sname	location	pno	pname	part_spec
S1	P1	150	S1	M/s Royal furniture	Delhi	P1	Table	Wood
S1	P2	50	S2	M/s Balaji furniture	Bangalore	P2	Chair	Wood
S1	P3	100	S3	M/s Premium furniture	Chennai	P3	Table	Steel
S2	P4	200				P4	Almirah	Steel
S2	P5	250				P5	Almirah	Wood
S3	P1	250						
S3	P2	150						
S3	P5	300						
S3	P4	250						

The primary key of each table is indicated by underlining the constituent fields.

```
SELECT s.sno, s.sname
FROM Suppliers s, Catalogue c
WHERE s.sno=c.sno AND
      cost > (SELECT AVG (cost)
              FROM Catalogue
              WHERE pno = 'P4'
              GROUP BY pno) ;
```

The number of rows returned by the above SQL query is

A. 4

B. 5

C. 0

D. 2

Answer key

3.21.43 SQL: GATE CSE 2021 | Set 1 | Question: 23



A relation $r(A, B)$ in a relational database has 1200 tuples. The attribute A has integer values ranging from 6 to 20, and the attribute B has integer values ranging from 1 to 20. Assume that the attributes A and B are independently distributed.

The estimated number of tuples in the output of $\sigma_{(A>10) \vee (B=18)}(r)$ is _____.

gatecse-2021-set1 databases sql numerical-answers one-mark

Answer key

3.21.44 SQL: GATE CSE 2021 | Set 2 | Question: 31



The relation scheme given below is used to store information about the employees of a company, where **empId** is the key and **deptId** indicates the department to which the employee is assigned. Each employee is assigned to exactly one department.

emp(empId, name, gender, salary, deptId)

Consider the following SQL query:

```
select deptId, count(*)
from emp
where gender = 'female' and salary > (select avg(salary) from emp)
group by deptId;
```

The above query gives, for each department in the company, the number of female employees whose salary is greater than the average salary of

- A. employees in the department
- B. employees in the company
- C. female employees in the department
- D. female employees in the company

gatecse-2021-set2 databases sql easy two-marks

Answer key

3.21.45 SQL: GATE CSE 2022 | Question: 46



Consider the relational database with the following four schemas and their respective instances.

- Student(sNo, sName, dNo) Dept(dNo, dName)
- Course(cNo, cName, dNo) Register(sNo, cNo)

Students		
sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Depth	
dNo	dName
D01	CSE
D02	EEE

Course		
cNo	cName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

SQL query

```
SELECT * FROM Student AS S WHERE NOT EXIST
```

```
(SELECT cNo FROM Course WHERE dNo = "D01"
```

```
EXCEPT
```

```
SELECT cNo FROM Register WHERE sNo = S.sNo)
```

The number of rows returned by the above SQL query is _____.

gatecse-2022 numerical-answers databases sql two-marks

Answer key

3.21.46 SQL: GATE CSE 2023 Question: 51



Consider the following table named Student in a relational database. The primary key of this table is rollNum.

Student

rollNum	name	gender	marks
1	Naman	M	62
2	Aliya	F	70
3	Aliya	F	80
4	James	M	82
5	Swati	F	65

The SQL query below is executed on this database.

```
SELECT *  
FROM Student  
WHERE gender = 'F' AND  
marks > 65
```

The number of rows returned by the query is _____.

gatecse-2023 databases sql numerical-answers two-marks easy

Answer key

3.21.47 SQL: GATE CSE 2025 Set 1 Question: 45



Consider the following database tables of a sports league.

```
player (pid, pname, age)    team (tid, tname, city, cid)  
coach (cid, cname)         members (pid, tid)
```

An instance of the table and an SQL query are given.

player

pid	pname	age
1	Jaspri	31
2	Atharva	24
3	Ishan	26
4	Axar	30

coach

cid	cname
101	Ricky
102	Mark
103	Trevor

team

tid	tname	city	cid
10	MI	Mumbai	102
20	DC	Delhi	101
30	PK	Mohali	103

members

pid	tid
1	10
2	30
3	10
4	20

```
SELECT MIN (P.age)  
FROM player P  
WHERE P.pid IN (  
    SELECT M.pid  
    FROM team T, coach C, members M  
    WHERE C.cname = 'Mark'  
    AND T.cid = C.cid  
    AND M.tid = T.tid  
)
```

The value returned by the given SQL query is _____. (Answer in integer)

Answer key

3.21.48 SQL: GATE CSE 2025 | Set 2 | Question: 44



Consider the following relational schema:

Students (rollno: integer, name: string, age: integer, cgpa: real)

Courses (courseno: integer, cname: string, credits: integer)

Enrolled (rollno: integer, courseno: integer, grade: string)

Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e., courseno) 1470?

A. SELECT S.name
FROM Students S
WHERE EXISTS (SELECT * FROM Enrolled E
WHERE E.courseno = 1470
AND E.rollno = S.rollno);

B. SELECT S.name
FROM Students S
WHERE SIZEOF (SELECT * FROM Enrolled E
WHERE E.courseno = 1470
AND E.rollno = S.rollno) > 0;

C. SELECT S.name
FROM Students S
WHERE 0 < (SELECT COUNT(*)
FROM Enrolled E
WHERE E.courseno = 1470
AND E.rollno = S.rollno);

D. SELECT S.name
FROM Students S NATURAL JOIN Enrolled E
WHERE E.courseno = 1470;

Answer key

3.21.49 SQL: GATE DA 2025 | Question: 23



On a relation named Loan of a bank:

Loan		
loan_number	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

the following SQL query is executed

```
SELECT L1.loan_number
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
FROM Loan L2
WHERE L2.branch_name = 'SR Nagar');
```

The number of rows returned by the query is _____ (Answer in integer).

Answer key

3.21.50 SQL: GATE DS&AI 2024 | Question: 21



Consider the following two tables named Raider and Team in a relational database maintained by a Kabaddi league. The attribute ID in table Team references the primary key of the Raider table, ID.

Raider				Team		
ID	Name	Raids	Raidpoints	City	ID	BidPoints
1	Arjun	200	250	Jaipur	2	200
2	Ankush	190	219	Patna	3	195
3	Sunil	150	200	Hyderabad	5	175
4	Reza	150	190	Jaipur	1	250
5	Pratham	175	220	Patna	4	200
6	Gopal	193	215	Jaipur	6	200

The SQL query described below is executed on this database:

```
SELECT *
FROM Raider, Team
WHERE Raider.ID = Team.ID AND City = "Jaipur" AND
RaidPoints > 200
```

The number of rows returned by this query is _____.

Answer key

3.21.51 SQL: GATE DS&AI 2024 | Question: 45



An OTT company is maintaining a large disk-based relational database of different movies with the following schema:

Movie (ID, CustomerRating)
 Genre (ID, Name)
 Movie_Genre (MovieID, GenreID)

Consider the following SQL query on the relation database above:

```
SELECT *
FROM Movie, Genre, Movie_Genre
WHERE
  Movie.CustomerRating > 3.4 AND
  Genre.Name = "Comedy" AND
  Movie_Genre.MovieID = Movie.ID AND
  Movie_Genre.GenreID = Genre.ID;
```

This SQL query can be sped up using which of the following indexing options?

- A. B⁺ tree on all the attributes.
- B. Hash index on Genre.Name and B⁺ tree on the remaining attributes.
- C. Hash index on Movie.CustomerRating and B⁺ tree on the remaining attributes.
- D. Hash index on all the attributes.

Answer key

3.21.52 SQL: GATE IT 2004 | Question: 74



A relational database contains two tables student and department in which student table has columns roll_no, name and dept_id and department table has columns dept_id and dept_name. The following insert statements were executed successfully to populate the empty tables:

```
Insert into department values (1, 'Mathematics')
Insert into department values (2, 'Physics')
Insert into student values (1, 'Navin', 1)
Insert into student values (2, 'Mukesh', 2)
Insert into student values (3, 'Gita', 1)
```

How many rows and columns will be retrieved by the following SQL statement?

```
Select * from student, department
```

- A. 0 row and 4 columns
B. 3 rows and 4 columns
C. 3 rows and 5 columns
D. 6 rows and 5 columns

gateit-2004 databases sql normal

Answer key

3.21.53 SQL: GATE IT 2004 | Question: 76



A table T1 in a relational database has the following rows and columns:

Roll no.	Marks
1	10
2	20
3	30
4	NULL

The following sequence of SQL statements was successfully executed on table T1.

```
Update T1 set marks = marks + 5
Select avg(marks) from T1
```

What is the output of the select statement?

- A. 18.75
B. 20
C. 25
D. Null

gateit-2004 databases sql normal

Answer key

3.21.54 SQL: GATE IT 2004 | Question: 78



Consider two tables in a relational database with columns and rows as follows:

Table: Student			Table: Department	
Roll_no	Name	Dept_id	Dept_id	Dept_name
1	ABC	1	1	A
2	DEF	1	2	B
3	GHI	2	3	C
4	JKL	3		

Roll_no is the primary key of the Student table. Dept_id is the primary key of the Department table and Student.Dept_id is a foreign key from Department.Dept_id

What will happen if we try to execute the following two SQL statements?

- i. update Student set Dept_id = Null where Roll_no = 1
ii. update Department set Dept_id = Null where Dept_id = 1

- A. Both i and ii will fail
B. i will fail but ii will succeed

C. i will succeed but ii will fail

D. Both i and ii will succeed

gateit-2004 databases sql normal

Answer key

3.21.55 SQL: GATE IT 2005 | Question: 69



In an inventory management system implemented at a trading corporation, there are several tables designed to hold all the information. Amongst these, the following two tables hold information on which items are supplied by which suppliers, and which warehouse keeps which items along with the stock-level of these items.

Supply = (supplierid, itemcode)

Inventory = (itemcode, warehouse, stocklevel)

For a specific information required by the management, following SQL query has been written

```
Select distinct STMP.supplierid
From Supply as STMP
Where not unique (Select ITMP.supplierid
                  From Inventory, Supply as ITMP
                  Where STMP.supplierid = ITMP.supplierid
                  And ITMP.itemcode = Inventory.itemcode
                  And Inventory.warehouse = 'Nagpur');
```

For the warehouse at Nagpur, this query will find all suppliers who

A. do not supply any item

B. supply exactly one item

C. supply one or more items

D. supply two or more items

gateit-2005 databases sql normal

Answer key

3.21.56 SQL: GATE IT 2006 | Question: 84



Consider a database with three relation instances shown below. The primary keys for the Drivers and Cars relation are *did* and *cid* respectively and the records are stored in ascending order of these primary keys as given in the tables. No indexing is available in the database.

D: Drivers relation

did	dname	rating	age
22	Karthikeyan	7	25
29	Salman	1	33
31	Boris	8	55
32	Amoldt	8	25
58	Schumacher	10	35
64	Sachin	7	35
71	Senna	10	16
74	Sachin	9	35
85	Rahul	3	25
95	Ralph	3	53

R: Reserves relation

did	Cid	day
22	101	10 / 10 / 06
22	102	10 / 10 / 06
22	103	08 / 10 / 06
22	104	07 / 10 / 06
31	102	10 / 11 / 16
31	103	06 / 11 / 16
31	104	12 / 11 / 16
64	101	05 / 09 / 06
64	102	08 / 09 / 06
74	103	08 / 09 / 06

C: Cars relation

Cid	Cname	colour
101	Renault	blue
102	Renault	red
103	Ferrari	green
104	Jaguar	red

What is the output of the following SQL query?

```
select D.dname
from Drivers D
where D.did in (
    select R.did
    from Cars C, Reserves R
    where R.cid = C.cid and C.colour = 'red'
    intersect
    select R.did
    from Cars C, Reserves R
    where R.cid = C.cid and C.colour = 'green'
)
```

- A. Karthikeyan, Boris
C. Karthikeyan, Boris, Sachin

- B. Sachin, Salman
D. Schumacher, Senna

gateit-2006 databases sql normal

Answer key

3.21.57 SQL: GATE IT 2006 | Question: 85



Consider a database with three relation instances shown below. The primary keys for the Drivers and Cars relation are *did* and *cid* respectively and the records are stored in ascending order of these primary keys as given in the tables. No indexing is available in the database.

D: Drivers relation

did	dname	rating	age
22	Karthikeyan	7	25
29	Salman	1	33
31	Boris	8	55
32	Amoldt	8	25
58	Schumacher	10	35
64	Sachin	7	35
71	Senna	10	16
74	Sachin	9	35
85	Rahul	3	25
95	Ralph	3	53

R: Reserves relation

did	Cid	day
22	101	10 – 10 – 06
22	102	10 – 10 – 06
22	103	08 – 10 – 06
22	104	07 – 10 – 06
31	102	10 – 11 – 16
31	103	06 – 11 – 16
31	104	12 – 11 – 16
64	101	05 – 09 – 06
64	102	08 – 09 – 06
74	103	08 – 09 – 06

C: Cars relation

Cid	Cname	colour
101	Renault	blue
102	Renault	red
103	Ferrari	green
104	Jaguar	red

```
select D.dname
from Drivers D
where D.did in (
    select R.did
```

```

from Cars C, Reserves R
where R.cid = C.cid and C.colour = 'red'
intersect
select R.cid
from Cars C, Reserves R
where R.cid = C.cid and C.colour = 'green'
)

```

Let n be the number of comparisons performed when the above SQL query is optimally executed. If linear search is used to locate a tuple in a relation using primary key, then n lies in the range:

- A. 36 – 40 B. 44 – 48 C. 60 – 64 D. 100 – 104

gateit-2006 databases sql normal

Answer key

3.21.58 SQL: GATE IT 2008 | Question: 74



Consider the following relational schema:

- Student(school-id, sch-roll-no, sname, saddress)
- School(school-id, sch-name, sch-address, sch-phone)
- Enrolment(school-id, sch-roll-no, erollno, examname)
- ExamResult(erollno, examname, marks)

What does the following SQL query output?

```

SELECT sch-name, COUNT (*)
FROM School C, Enrolment E, ExamResult R
WHERE E.school-id = C.school-id
AND
E.examname = R.examname AND E.erollno = R.erollno
AND
R.marks = 100 AND E.school-id IN (SELECT school-id
FROM student
GROUP BY school-id
HAVING COUNT (*) > 200)
GROUP BY school-id

```

- A. for each school with more than 200 students appearing in exams, the name of the school and the number of 100s scored by its students
- B. for each school with more than 200 students in it, the name of the school and the number of 100s scored by its students
- C. for each school with more than 200 students in it, the name of the school and the number of its students scoring 100 in at least one exam
- D. nothing; the query has a syntax error

gateit-2008 databases sql normal

Answer key

3.22

Safe Query (1)

3.22.1 Safe Query: GATE CSE 2017 | Set 1 | Question: 41



Consider a database that has the relation schemas EMP(Empld, EmpName, Deptld), and DEPT(DeptName, Deptld). Note that the Deptld can be permitted to be NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

- $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \forall v \in \text{DEPT}(t[\text{Deptld}] \neq v[\text{Deptld}]))\}$
- $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{Deptld}] \neq v[\text{Deptld}]))\}$
- $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{Deptld}] = v[\text{Deptld}]))\}$

Which of the above queries are safe?

- A. I and II only B. I and III only C. II and III only D. I, II and III

Answer key

3.23

Super Key (1)

3.23.1 Super Key: GATE CSE 2026 | Set 1 | Question: 55

Consider a relational database schema with a relation $R(A, B, C, D)$. If $\{A, B\}$ and $\{A, C\}$ are the only two candidate keys of the relation R , then the number of superkeys of relation R is _____. (answer in integer)

gatecse-2026-set1 numerical-answers databases super-key two-marks

Answer key

3.24

Timestamp Ordering (1)

3.24.1 Timestamp Ordering: GATE CSE 2017 | Set 1 | Question: 42

In a database system, unique timestamps are assigned to each transaction using Lamport's logical clock. Let $TS(T_1)$ and $TS(T_2)$ be the timestamps of transactions T_1 and T_2 respectively. Besides, T_1 holds a lock on the resource R , and T_2 has requested a conflicting lock on the same resource R . The following algorithm is used to prevent deadlocks in the database system assuming that a killed transaction is restarted with the same timestamp.

if $TS(T_2) < TS(T_1)$ then

T_1 is killed

else T_2 waits.

Assume any transaction that is not killed terminates eventually. Which of the following is TRUE about the database system that uses the above algorithm to prevent deadlocks?

- A. The database system is both deadlock-free and starvation-free.
- B. The database system is deadlock-free, but not starvation-free.
- C. The database system is starvation-free, but not deadlock-free.
- D. The database system is neither deadlock-free nor starvation-free.

gatecse-2017-set1 databases timestamp-ordering normal transaction-and-concurrency

Answer key

3.25

Transaction and Concurrency (27)

 **Practice Tests:** [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(11Q\)](#)

3.25.1 Transaction and Concurrency: GATE CS Practice : Transaction Concurrency Control (DBMS)

We have a table Orders with a specific predicate range query

Q: `SELECT * FROM Orders WHERE Value > 1000`

Transaction T_1 executes Q and gets 5 rows.

Transaction T_2 inserts a new row with Value = 1500 and Commits.

Transaction T_1 executes Q again.

Under which Isolation Level is it **guaranteed** that T_1 will still see exactly 5 rows (i.e., prevent the Phantom Read)?

- A. Read Committed
- B. Repeatable Read
- C. Serializable
- D. Both B and C

transaction-and-concurrency databases gate-preparation

3.25.2 Transaction and Concurrency: GATE CSE 1999 | Question: 2.6



For the schedule given below, which of the following is correct:

- | | |
|---|---------|
| 1 | Read A |
| 2 | Read B |
| 3 | Write A |
| 4 | Read A |
| 5 | Write A |
| 6 | Write B |
| 7 | Read B |
| 8 | Write B |

- A. This schedule is serializable and can occur in a scheme using 2PL protocol
 B. This schedule is serializable but cannot occur in a scheme using 2PL protocol
 C. This schedule is not serializable but can occur in a scheme using 2PL protocol
 D. This schedule is not serializable and cannot occur in a scheme using 2PL protocol

gate1999 databases transaction-and-concurrency normal

3.25.3 Transaction and Concurrency: GATE CSE 2003 | Question: 29, ISRO2009-73



Which of the following scenarios may lead to an irrecoverable error in a database system?

- A. A transaction writes a data item after it is read by an uncommitted transaction
 B. A transaction reads a data item after it is read by an uncommitted transaction
 C. A transaction reads a data item after it is written by a committed transaction
 D. A transaction reads a data item after it is written by an uncommitted transaction

gatecse-2003 databases transaction-and-concurrency easy isro2009

3.25.4 Transaction and Concurrency: GATE CSE 2003 | Question: 87



Consider three data items D_1 , D_2 , and D_3 , and the following execution schedule of transactions T_1 , T_2 , and T_3 . In the diagram, $R(D)$ and $W(D)$ denote the actions reading and writing the data item D respectively.

T1	T2	T3
R(D1); W(D1);	R(D3); R(D2); W(D2);	R(D2); R(D3);
R(D2); W(D2);	R(D1); W(D1);	W(D2); W(D3);

Which of the following statements is correct?

- A. The schedule is serializable as $T2; T3; T1$
- B. The schedule is serializable as $T2; T1; T3$
- C. The schedule is serializable as $T3; T2; T1$
- D. The schedule is not serializable

gatecse-2003 databases transaction-and-concurrency normal

Answer key

3.25.5 Transaction and Concurrency: GATE CSE 2006 | Question: 20, ISRO2015-17

Consider the following log sequence of two transactions on a bank account, with initial balance 12000, that transfer 2000 to a mortgage payment and then apply a 5% interest.

1. T1 start
2. T1 B_{old} = 12000 new = 10000
3. T1 M_{old} = 0 new = 2000
4. T1 commit
5. T2 start
6. T2 B_{old} = 10000 new = 10500
7. T2 commit

Suppose the database system crashes just before log record 7 is written. When the system is restarted, which one statement is true of the recovery procedure?

- A. We must redo log record 6 to set B to 10500
- B. We must undo log record 6 to set B to 10000 and then redo log records 2 and 3
- C. We need not redo log records 2 and 3 because transaction T1 has committed
- D. We can apply redo and undo operations in arbitrary order because they are idempotent

gatecse-2006 databases transaction-and-concurrency normal isro2015

Answer key

3.25.6 Transaction and Concurrency: GATE CSE 2010 | Question: 20

Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?

- I. 2-phase locking
II. Time-stamp ordering

A. I only

B. II only

C. Both I and II

D. Neither I nor II

gatecse-2010 databases transaction-and-concurrency normal

Answer key

3.25.7 Transaction and Concurrency: GATE CSE 2010 | Question: 42



Consider the following schedule for transactions T_1 , T_2 and T_3 :

T1	T2	T3
Read(X)		
	Read(Y)	
		Read(Y)
	Write(Y)	
Write(X)		
		Write(X)
	Read(X)	
	Write(X)	

Which one of the schedules below is the correct serialization of the above?

A. $T_1 \rightarrow T_3 \rightarrow T_2$

B. $T_2 \rightarrow T_1 \rightarrow T_3$

C. $T_2 \rightarrow T_3 \rightarrow T_1$

D. $T_3 \rightarrow T_1 \rightarrow T_2$

gatecse-2010 databases transaction-and-concurrency normal

Answer key

3.25.8 Transaction and Concurrency: GATE CSE 2012 | Question: 27



Consider the following transactions with data items P and Q initialized to zero:

T_1	read (P); read (Q); if $P = 0$ then $Q := Q + 1$; write (Q)
T_2	read (Q); read (P); if $Q = 0$ then $P := P + 1$; write (P)

Any non-serial interleaving of T_1 and T_2 for concurrent execution leads to

A. a serializable schedule

B. a schedule that is not conflict serializable

C. a conflict serializable schedule

D. a schedule for which a precedence graph cannot be drawn

gatecse-2012 databases transaction-and-concurrency normal

Answer key

3.25.9 Transaction and Concurrency: GATE CSE 2015 | Set 2 | Question: 1



Consider the following transaction involving two bank accounts x and y .

read(x); x:=x-50; write(x); read(y); y:=y+50; write(y)

The constraint that the sum of the accounts x and y should remain constant is that of

- A. Atomicity B. Consistency C. Isolation D. Durability

gatecse-2015-set2 databases transaction-and-concurrency easy

Answer key

3.25.10 Transaction and Concurrency: GATE CSE 2015 | Set 2 | Question: 46



Consider a simple checkpointing protocol and the following set of operations in the log.

(start, T4); (write, T4, y, 2, 3); (start, T1); (commit, T4); (write, T1, z, 5, 7);

(checkpoint);

(start, T2); (write, T2, x, 1, 9); (commit, T2); (start, T3); (write, T3, z, 7, 2);

If a crash happens now and the system tries to recover using both undo and redo operations, what are the contents of the undo list and the redo list?

- A. Undo: T3, T1; Redo: T2 B. Undo: T3, T1; Redo: T2, T4
C. Undo: none; Redo: T2, T4, T3, T1 D. Undo: T3, T1, T4; Redo: T2

gatecse-2015-set2 databases transaction-and-concurrency normal

Answer key

3.25.11 Transaction and Concurrency: GATE CSE 2015 | Set 3 | Question: 29



Consider the partial Schedule S involving two transactions T_1 and T_2 . Only the read and the write operations have been shown. The read operation on data item P is denoted by $read(P)$ and write operation on data item P is denoted by $write(P)$.

Schedule S		
Time Instance	Transaction ID	
	T1	T2
1	read(A)	
2	write(A)	
3		read(C)
4		write(C)
5		read(B)
6		write(B)
7		read(A)
8		commit
9	read(B)	

Suppose that the transaction T_1 fails immediately after time instance 9. Which of the following statements is correct?

- A. T_2 must be aborted and then both T_1 and T_2 must be re-started to ensure transaction atomicity
B. Schedule S is non-recoverable and cannot ensure transaction atomicity
C. Only T_2 must be aborted and then re-started to ensure transaction atomicity
D. Schedule S is recoverable and can ensure transaction atomicity and nothing else needs to be done

gatecse-2015-set3 databases transaction-and-concurrency normal

Answer key

3.25.12 Transaction and Concurrency: GATE CSE 2016 | Set 1 | Question: 22



Which one of the following is NOT a part of the ACID properties of database transactions?

- A. Atomicity B. Consistency C. Isolation D. Deadlock-freedom

gatecse-2016-set1 databases transaction-and-concurrency easy

Answer key

3.25.13 Transaction and Concurrency: GATE CSE 2016 | Set 1 | Question: 51



Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{O_1, \dots, O_k\}$. This is done in the following manner:

- Step 1. T acquires exclusive locks to $O_1, \dots, O_k$ in increasing order of their addresses.
- Step 2. The required operations are performed.
- Step 3. All locks are released.

This protocol will

- A. guarantee serializability and deadlock-freedom
 B. guarantee neither serializability nor deadlock-freedom
 C. guarantee serializability but not deadlock-freedom
 D. guarantee deadlock-freedom but not serializability

gatecse-2016-set1 databases transaction-and-concurrency normal

Answer key

3.25.14 Transaction and Concurrency: GATE CSE 2016 | Set 2 | Question: 22



Suppose a database schedule S involves transactions $T_1, \dots, T_n$. Construct the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?

- A. Topological order B. Depth-first order
 C. Breadth-first order D. Ascending order of the transaction indices

gatecse-2016-set2 databases transaction-and-concurrency normal

Answer key

3.25.15 Transaction and Concurrency: GATE CSE 2016 | Set 2 | Question: 51



Consider the following database schedule with two transactions T_1 and T_2 .

$S = r_2(X); r_1(X); r_2(Y); w_1(X); r_1(Y); w_2(X); a_1; a_2$

Where $r_i(Z)$ denotes a read operation by transaction T_i on a variable Z , $w_i(Z)$ denotes a write operation by T_i on a variable Z and a_i denotes an abort by transaction T_i .

Which one of the following statements about the above schedule is TRUE?

- A. S is non-recoverable. B. S is recoverable, but has a cascading abort.
 C. S does not have a cascading abort. D. S is strict.

gatecse-2016-set2 databases transaction-and-concurrency normal

Answer key

3.25.16 Transaction and Concurrency: GATE CSE 2019 | Question: 11



Consider the following two statements about database transaction schedules:

- I. Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable.
 II. Timestamp-ordering concurrency control protocol with Thomas' Write Rule can generate view serializable schedules that are not conflict serializable

Which of the above statements is/are TRUE?

- A. I only B. II only C. Both I and II D. Neither I nor II

Answer key

3.25.17 Transaction and Concurrency: GATE CSE 2021 | Set 1 | Question: 13



Suppose a database system crashes again while recovering from a previous crash. Assume checkpointing is not done by the database either during the transactions or during recovery.

Which of the following statements is/are correct?

- A. The same undo and redo list will be used while recovering again
- B. The system cannot recover any further
- C. All the transactions that are already undone and redone will not be recovered again
- D. The database will become inconsistent

Answer key

3.25.18 Transaction and Concurrency: GATE CSE 2024 | Set 2 | Question: 9



Once the DBMS informs the user that a transaction has been successfully completed, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called

- A. durability
- B. atomicity
- C. consistency
- D. isolation

Answer key

3.25.19 Transaction and Concurrency: GATE CSE 2025 | Set 1 | Question: 5



A schedule of three database transactions T_1, T_2 , and T_3 is shown. $R_i(A)$ and $W_i(A)$ denote read and write of data item A by transaction $T_i, i = 1, 2, 3$. The transaction T_1 aborts at the end. Which other transaction(s) will be required to be rolled back?

$$R_1(X)W_1(Y)R_2(X)R_2(Y)R_3(Y) \text{ ABORT}(T_1)$$

- A. Only T_2
- B. Only T_3
- C. Both T_2 and T_3
- D. Neither T_2 nor T_3

Answer key

3.25.20 Transaction and Concurrency: GATE CSE 2025 | Set 2 | Question: 17



An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account A attempted simultaneous transfers of Rs. 10000 each from account A to account B . Both transactions read the same value, Rs. 11000, as the initial balance in A and were allowed to go through. B was credited Rs. 10000 twice. A was debited only once and ended up with a balance of Rs. 1000.

Which of the following properties is/are certain to have been violated by the system?

- A. Atomicity
- B. Consistency
- C. Isolation
- D. Durability

Answer key

3.25.21 Transaction and Concurrency: GATE CSE 2026 | Set 2 | Question: 10



Consider concurrent execution of two transactions T_1 and T_2 in a DBMS, both of which access a data object A . For these two transactions to not conflict on A , which one of the following statements must be true?

- A. Both T_1 and T_2 only read A
- B. T_1 reads A and T_2 writes A
- C. T_1 writes A and T_2 reads A
- D. Both T_1 and T_2 write A

Answer key

3.25.22 Transaction and Concurrency: GATE IT 2004 | Question: 21



Which level of locking provides the highest degree of concurrency in a relational database?

- A. Page
- B. Table
- C. Row
- D. Page, table and row level locking allow the same degree of concurrency

gateit-2004 databases normal transaction-and-concurrency

Answer key

3.25.23 Transaction and Concurrency: GATE IT 2004 | Question: 77



Consider the following schedule S of transactions T_1 and T_2 :

T_1	T_2
Read(A) $A = A - 10$	Read(A) $Temp = 0.2 * A$ Write(A) Read(B)
Write(A) Read(B) $B = B + 10$ Write(B)	$B = B + Temp$ Write(B)

Which of the following is TRUE about the schedule S ?

- A. S is serializable only as T_1, T_2
- B. S is serializable only as T_2, T_1
- C. S is serializable both as T_1, T_2 and T_2, T_1
- D. S is not serializable either as T_1, T_2 or as T_2, T_1

gateit-2004 databases transaction-and-concurrency normal

Answer key

3.25.24 Transaction and Concurrency: GATE IT 2005 | Question: 24



Amongst the ACID properties of a transaction, the 'Durability' property requires that the changes made to the database by a successful transaction persist

- A. Except in case of an Operating System crash
- B. Except in case of a Disk crash
- C. Except in case of a power failure
- D. Always, even if there is a failure of any kind

gateit-2005 databases transaction-and-concurrency easy

Answer key

3.25.25 Transaction and Concurrency: GATE IT 2005 | Question: 67



A company maintains records of sales made by its salespersons and pays them commission based on each individual's total sales made in a year. This data is maintained in a table with following schema:

salesinfo = (salespersonid, totalsales, commission)

In a certain year, due to better business results, the company decides to further reward its salespersons by enhancing the commission paid to them as per the following formula:

If commission < 50000, enhance it by 2%

If 50000 < commission < 100000, enhance it by 4%

If commission > 100000, enhance it by 6%

The IT staff has written three different SQL scripts to calculate enhancement for each slab, each of these scripts is to run as a separate transaction as follows:

T1 Update salesinfo
Set commission = commission * 1.02
Where commission <= 50000;

T2 Update salesinfo
Set commission = commission * 1.04
Where commission > 50000 and
commission <= 100000;

T3 Update salesinfo
Set commission = commission * 1.06
Where commission > 100000;

Which of the following options of running these transactions will update the commission of all salespersons correctly

- A. Execute T1 followed by T2 followed by T3
- B. Execute T2, followed by T3; T1 running concurrently throughout
- C. Execute T3 followed by T2; T1 running concurrently throughout
- D. Execute T3 followed by T2 followed by T1

gateit-2005 databases transaction-and-concurrency normal

Answer key

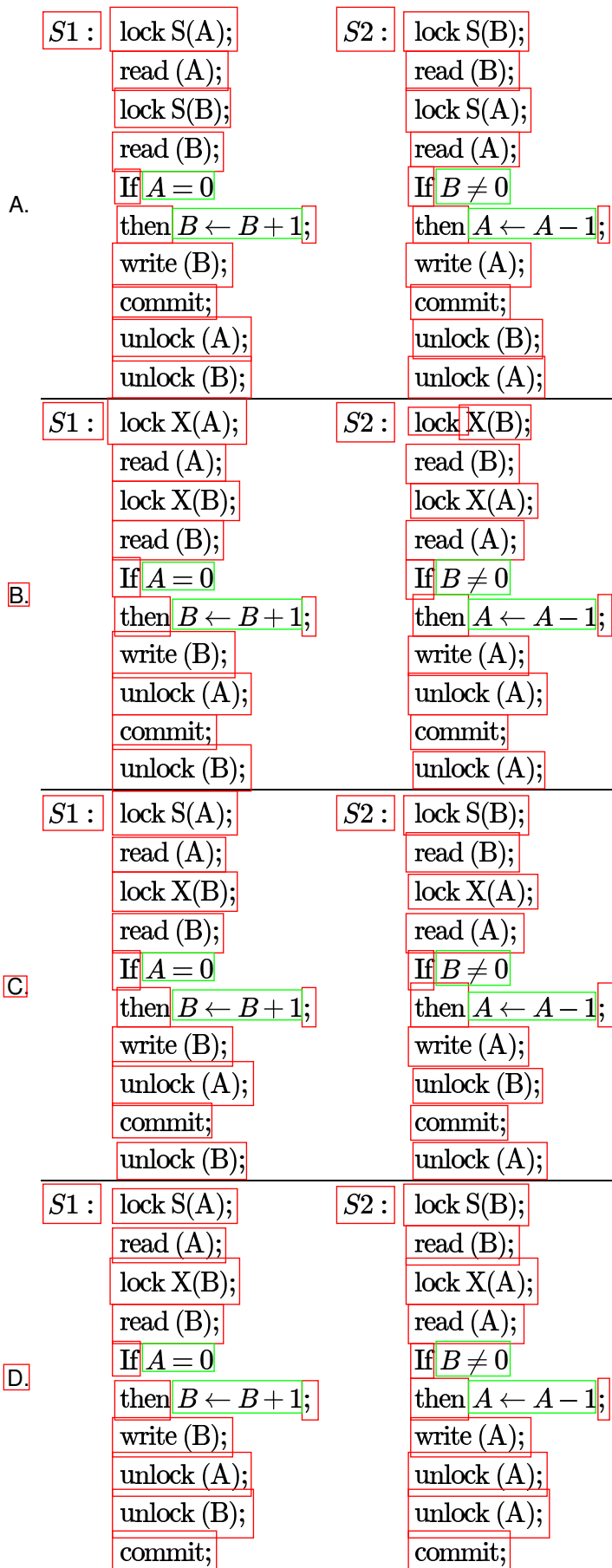
3.25.26 Transaction and Concurrency: GATE IT 2007 | Question: 66



Consider the following two transactions: T1 and T2.

T1 : read (A); read (B); If A = 0 then B ← B + 1; write (B);	T2 : read (B); read (A); If B ≠ 0 then A ← A - 1; write (A);
---	---

Which of the following schemes, using shared and exclusive locks, satisfy the requirements for strict two phase locking for the above transactions?



3.25.27 Transaction and Concurrency: GATE IT 2008 | Question: 63



Consider the following three schedules of transactions T1, T2 and T3. [Notation: In the following NYO represents the action Y (R for read, W for write) performed by transaction N on object O.]

(S1)	2RA	2WA	3RC	2WB	3WA	3WC	1RA	1RB	1WA	1WB
(S2)	3RC	2RA	2WA	2WB	3WA	1RA	1RB	1WA	1WB	3WC
(S3)	2RA	3RC	3WA	2WA	2WB	3WC	1RA	1RB	1WA	1WB

Which of the following statements is TRUE?

- A. S1, S2 and S3 are all conflict equivalent to each other
- B. No two of S1, S2 and S3 are conflict equivalent to each other
- C. S2 is conflict equivalent to S3, but not to S1
- D. S1 is conflict equivalent to S2, but not to S3

gateit-2008 databases transaction-and-concurrency normal

Answer key

3.26 Tuple Relational Calculus (3)

3.26.1 Tuple Relational Calculus: GATE CSE 2025 | Set 1 | Question: 29



Consider two relations describing teams and players in a sports league:

- teams(tid, tname): tid, tname are team-id and team-name, respectively
- players(pid, pname, tid): pid, pname, and tid denote player-id, playername and the team-id of the player, respectively

Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having tname as 'MI'?

- A. $\{p.pname \mid p \in \text{players} \wedge \exists t (t \in \text{teams} \wedge p.tid = t.tid \wedge t.tname = 'MI')\}$
- B. $\{p.pname \mid p \in \text{teams} \wedge \exists t (t \in \text{players} \wedge p.tid = t.tid \wedge t.tname = 'MI')\}$
- C. $\{p.pname \mid p \in \text{players} \wedge \exists t (t \in \text{teams} \wedge t.tname = 'MI')\}$
- D. $\{p.pname \mid p \in \text{teams} \wedge \exists t (t \in \text{players} \wedge t.tname = 'MI')\}$

gatecse2025-set1 databases tuple-relational-calculus two-marks

Answer key

3.26.2 Tuple Relational Calculus: GATE CSE 2026 | Set 1 | Question: 33



Consider a relational database schema with two relations $R(P, Q)$ and $S(X, Y)$.

Let $E = \{\langle u \rangle \mid \exists v \exists w \langle u, v \rangle \in R \wedge \langle v, w \rangle \in S\}$ be a tuple relational calculus expression.

Which one of the following relational algebraic expressions is equivalent to E ?

- A. $\Pi_P (R \bowtie_{R.P=S.X} S)$
- B. $\Pi_P (S \bowtie_{S.X=R.Q} R)$
- C. $\Pi_P (R \bowtie_{R.P=S.Y} S)$
- D. $\Pi_P (S \bowtie_{S.Y=R.Q} R)$

gatecse-2026-set1 databases two-marks tuple-relational-calculus

Answer key

3.26.3 Tuple Relational Calculus: GATE DA 2026 | Question: 49



Let there be two relations X and Y as shown. X has three columns P, Q and R . Y has two columns P and S .

X

P	Q	R
P1	Q1	R1
P2	Q2	R2
P3	Q3	R2

Y

P	S
P1	10
P1	15
P2	20
P3	1

Consider that the following tuple relational calculus expression is evaluated.

$$\{t \mid t \in X \wedge \exists z \in X (t[P] = z[P]) \wedge \exists m \in Y (m[P] = t[P] \wedge m[S] > 1)\}$$

The number of tuples that will be returned is _____. (Answer in integer)

gateda-2026 databases tuple-relational-calculus numerical-answers two-marks

Answer key

3.27

Two Phase Locking Protocol (1)

3.27.1 Two Phase Locking Protocol: GATE CSE 2024 | Set 2 | Question: 17



Which of the following statements about the Two Phase Locking (2PL) protocol is/are TRUE?

- A. 2PL permits only serializable schedules
- B. With 2PL, a transaction always locks the data item being read or written just before every operation and always releases the lock just after the operation
- C. With 2PL, once a lock is released on any data item inside a transaction, no more locks on any data item can be obtained inside that transaction
- D. A deadlock is possible with 2PL

gatecse-2024-set2 databases two-phase-locking-protocol multiple-selects transaction-and-concurrency one-mark

Answer key

Answer Keys

3.1.1	D	3.2.1	N/A	3.2.2	N/A	3.2.3	N/A	3.2.4	N/A
3.2.5	B	3.2.6	N/A	3.2.7	B	3.2.8	N/A	3.2.9	N/A
3.2.10	N/A	3.2.11	C	3.2.12	B	3.2.13	C	3.2.14	D
3.2.15	A	3.2.16	C	3.2.17	C	3.2.18	B	3.2.19	5
3.2.20	50	3.2.21	A	3.2.22	52	3.2.23	B	3.2.24	A
3.2.25	B,D	3.2.26	33:33	3.2.27	A	3.2.28	C	3.2.29	A
3.2.30	C	3.2.31	A	3.2.32	A	3.3.1	N/A	3.3.2	D
3.3.3	A	3.3.4	8	3.3.5	19	3.3.6	B	3.3.7	A

3.4.1	C	3.4.2	B	3.4.3	D	3.4.4	C	3.4.5	A
3.4.6	54	3.4.7	A	3.4.8	B	3.4.9	B	3.4.10	A
3.4.11	B;C;D	3.4.12	B	3.5.1	False	3.6.1	True	3.6.2	N/A
3.6.3	N/A	3.6.4	N/A	3.6.5	N/A	3.6.6	N/A	3.6.7	A;B;D
3.6.8	False	3.6.9	N/A	3.6.10	A	3.6.11	D	3.6.12	N/A
3.6.13	B	3.6.14	D	3.6.15	B	3.6.16	C	3.6.17	A
3.6.18	N/A	3.6.19	C	3.6.20	C	3.6.21	D	3.6.22	B
3.6.23	C	3.6.24	C	3.6.25	D	3.6.26	C	3.6.27	C
3.6.28	B	3.6.29	A	3.6.30	B	3.6.31	A	3.6.32	C
3.6.33	B	3.6.34	B	3.6.35	A	3.6.36	B	3.6.37	C
3.6.38	A	3.6.39	A	3.6.40	A;C;D	3.6.41	8	3.6.42	A
3.6.43	A;C	3.6.44	B;C;D	3.6.45	50	3.6.46	C;D	3.6.47	A;B;C
3.6.48	A;B;D	3.6.49	TBA	3.6.50	B	3.6.51	A	3.6.52	B
3.6.53	B	3.6.54	A	3.6.55	D	3.6.56	C	3.7.1	C
3.8.1	C	3.9.1	B	3.9.2	B	3.9.3	A	3.9.4	C
3.9.5	4	3.9.6	C	3.9.7	A	3.9.8	A	3.9.9	D
3.9.10	A	3.9.11	B	3.9.12	C	3.10.1	B;C	3.10.2	C;D
3.10.3	B;D	3.11.1	N/A	3.11.2	N/A	3.11.3	3	3.11.4	C
3.11.5	A	3.11.6	C	3.11.7	C	3.11.8	C	3.11.9	C
3.11.10	4	3.11.11	698 : 698	3.11.12	6	3.11.13	A;B	3.11.14	A
3.11.15	195:195	3.12.1	A	3.12.2	A	3.12.3	A	3.12.4	C
3.12.5	B	3.12.6	A	3.12.7	A	3.13.1	C	3.14.1	C
3.14.2	A	3.14.3	C	3.15.1	4:4	3.16.1	N/A	3.16.2	A
3.16.3	3:3	3.17.1	B	3.17.2	C	3.17.3	A	3.17.4	0.00
3.17.5	D	3.17.6	A;B	3.18.1	N/A	3.18.2	N/A	3.18.3	N/A
3.18.4	N/A	3.18.5	N/A	3.18.6	N/A	3.18.7	D	3.18.8	N/A
3.18.9	B	3.18.10	C	3.18.11	D	3.18.12	C	3.18.13	N/A
3.18.14	N/A	3.18.15	A	3.18.16	D	3.18.17	B	3.18.18	D
3.18.19	A	3.18.20	D	3.18.21	D	3.18.22	4	3.18.23	C
3.18.24	1	3.18.25	C	3.18.26	A;B	3.18.27	2	3.18.28	B
3.18.29	1:1	3.18.30	C	3.18.31	D	3.18.32	C	3.18.33	B
3.19.1	N/A	3.19.2	D	3.19.3	C	3.19.4	C	3.19.5	C
3.19.6	B	3.19.7	C	3.19.8	C	3.19.9	A	3.19.10	A
3.19.11	D	3.19.12	D	3.19.13	C	3.20.1	A	3.20.2	B;C
3.21.1	N/A	3.21.2	N/A	3.21.3	N/A	3.21.4	N/A	3.21.5	N/A
3.21.6	N/A	3.21.7	N/A	3.21.8	D	3.21.9	N/A	3.21.10	N/A
3.21.11	A	3.21.12	C	3.21.13	N/A	3.21.14	C	3.21.15	N/A
3.21.16	N/A	3.21.17	C	3.21.18	D	3.21.19	D	3.21.20	C
3.21.21	B	3.21.22	C	3.21.23	A	3.21.24	X	3.21.25	A
3.21.26	C	3.21.27	A	3.21.28	C	3.21.29	C	3.21.30	B

3.21.31	D	3.21.32	B
3.21.36	A	3.21.37	2
3.21.41	5	3.21.42	A
3.21.46	2	3.21.47	26:26
3.21.51	A;B	3.21.52	D
3.21.56	A	3.21.57	B
3.24.1	A	3.25.1	TBA
3.25.5	B	3.25.6	B
3.25.10	A	3.25.11	B
3.25.15	C	3.25.16	C
3.25.20	B;C	3.25.21	A
3.25.25	D	3.25.26	C
3.26.3	2	3.27.1	A;C;D

3.21.33	C	3.21.34	D	3.21.35	2
3.21.38	2.6	3.21.39	7	3.21.40	D
3.21.43	819 : 820 ; 205 : 205	3.21.44	B	3.21.45	2
3.21.48	A;C;D	3.21.49	3:3	3.21.50	3
3.21.53	C	3.21.54	C	3.21.55	D
3.21.58	D	3.22.1	D	3.23.1	6 : 6
3.25.2	D	3.25.3	D	3.25.4	D
3.25.7	A	3.25.8	B	3.25.9	B
3.25.12	D	3.25.13	A	3.25.14	A
3.25.17	A	3.25.18	A	3.25.19	C
3.25.22	C	3.25.23	X	3.25.24	D
3.25.27	D	3.26.1	A	3.26.2	B



Boolean algebra, Combinational and sequential circuits, Minimization, Number representations and computer arithmetic (fixed and floating point)

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	2	3	2	3	2	2	2	1	2	3	1
2 Marks Count	3	2	2	3	2	2	2	2	2	2	2
Total Marks	8	7	6	9	6	6	6	5	6	7	5

Welcome to the "Digital Logic" chapter of your GATE Computer Science exam preparation. This foundational subject is the bedrock of computer hardware, providing the essential understanding of how digital circuits process information, store data, and execute instructions. It delves into the fundamental building blocks of modern computing systems, from basic logic gates to complex sequential circuits and memory elements. A strong grasp of Digital Logic is crucial not only for direct questions in the GATE exam but also for understanding related concepts in Computer Architecture, Operating Systems, and even some aspects of programming. Typically, Digital Logic carries a weightage of 8-12 marks in the GATE CS paper, with questions ranging from conceptual understanding and circuit analysis to design problems involving minimization, number representation, and sequential circuit behavior. Questions often include Multiple Choice Questions (MCQs), Multiple Select Questions (MSQs), and Numerical Answer Type (NAT) questions, testing both theoretical knowledge and problem-solving skills.

Topic-wise Key Concepts

Adder

An adder is a digital circuit that performs the addition of numbers. It is a fundamental component in Arithmetic Logic Units (ALUs) of processors.

- **Half Adder (HA):** Adds two single-bit binary numbers. It produces a sum (S) and a carry-out (Cout).

◦ Formulas:

$$S = A \oplus B$$

$$C_{out} = A \cdot B$$

- **Full Adder (FA):** Adds three single-bit binary numbers (two input bits and a carry-in). It produces a sum (S) and a carry-out (Cout).

◦ Formulas:

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$$

$$C_{out} = (A \cdot B) + (B \cdot C_{in}) + (C_{in} \cdot A)$$

- **Ripple Carry Adder:** An n-bit adder constructed by cascading n full adders, where the carry-out of one stage becomes the carry-in of the next.

◦ Propagation Delay: The total delay is proportional to n, as carry ripples through all stages.

$$T_{ripple} = n \times T_{FA_carry}$$

- **Carry Look-Ahead Adder (CLA):** A faster adder that generates carries in parallel, reducing propagation delay. It uses carry generate (G) and carry propagate (P) signals.

◦ Formulas for a single stage i:

$$P_i = A_i \oplus B_i$$

$$G_i = A_i \cdot B_i$$

$$S_i = P_i \oplus C_i$$

$$C_{i+1} = G_i + (P_i \cdot C_i)$$

◦ For a 4-bit CLA:

$$C_1 = G_0 + P_0 C_0$$

$$C_2 = G_1 + P_1 C_1 = G_1 + P_1 G_0 + P_1 P_0 C_0$$

$$C_3 = G_2 + P_2 C_2 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$

$$C_4 = G_3 + P_3 C_3 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0$$

- **Common Pitfalls:** Confusing half adder and full adder functionality. Incorrectly calculating propagation delay for ripple carry vs. CLA.
- **Problem-solving:** Be able to draw and analyze basic adder circuits. Calculate delays for different adder types.

Array Multiplier

An array multiplier is a combinational circuit that performs multiplication of two binary numbers using an array of full adders and AND gates. It's a direct hardware implementation of the traditional "paper-and-pencil" multiplication method.

- **Core Idea:** For two n -bit numbers, it requires n rows of partial products. Each partial product is generated by ANDing the multiplicand with one bit of the multiplier. These partial products are then summed using an array of adders (often full adders) in a staggered fashion.
- **Structure:** An $n \times m$ bit array multiplier typically uses $n \cdot m$ AND gates to generate partial products and $(n-1) \cdot m$ full adders (or similar) to sum them.
- **Delay:** The delay is proportional to $n+m$, as carries propagate diagonally across the array.
- **Problem-solving:** Understand how partial products are formed and summed. Be able to trace the data flow for small examples.

Binary Codes

Binary codes are systems used to represent numbers, characters, and instructions in digital systems using binary digits (bits).

- **Weighted Codes:** Each bit position has a specific weight.
 - **BCD (8421):** Each decimal digit is represented by its 4-bit binary equivalent. E.g., $(9)_{10} = (1001)_{BCD}$. $(25)_{10} = (0010\ 0101)_{BCD}$.
 - **Excess-3:** BCD code with 3 added to each digit. Self-complementing.
 - **2421 Code:** Another weighted code.
- **Non-Weighted Codes:** Bit positions do not have fixed weights.
 - **Gray Code:** Only one bit changes between successive numbers. Used to prevent glitches in ADCs and K-maps.
 - **Binary to Gray:** $G_i = B_i \oplus B_{i+1}$ (for MSB to LSB, $G_{n-1} = B_{n-1}$, $G_i = B_i \oplus B_{i+1}$ for $i < n-1$). More commonly: $G_i = B_i \oplus B_{i+1}$, with $B_n = 0$ for the MSB. Correct: $G_{MSB} = B_{MSB}$. $G_i = B_i \oplus B_{i+1}$ for i from MSB-1 down to 0. Alternative: $G_i = B_i \oplus B_{i-1}$ for $i > 0$, $G_0 = B_0$. Example: Binary 1011 (11) $\rightarrow$ Gray: $G_3 = 1$, $G_2 = 1 \oplus 0 = 1$, $G_1 = 0 \oplus 1 = 1$, $G_0 = 1 \oplus 1 = 0$. So 1110.
 - **Gray to Binary:** $B_{MSB} = G_{MSB}$, $B_i = G_i \oplus B_{i+1}$ (for MSB to LSB). Correct: $B_i = G_i \oplus B_{i+1}$ for i from MSB-1 down to 0, $B_{MSB} = G_{MSB}$. Example: Gray 1110 $\rightarrow$ Binary: $B_3 = 1$, $B_2 = 1 \oplus 1 = 0$, $B_1 = 1 \oplus 0 = 1$, $B_0 = 0 \oplus 1 = 1$. So 1011.
 - **ASCII:** 7-bit or 8-bit code for characters.
- **Error Detection Codes (Parity):** Adds an extra bit to detect single-bit errors.
 - **Even Parity:** Total number of 1s (including parity bit) is even.
 - **Odd Parity:** Total number of 1s (including parity bit) is odd.
- **Error Correction Codes (Hamming Code):** Can detect and correct multiple-bit errors.
- **Common Pitfalls:** Confusing BCD with pure binary. Incorrectly converting between binary and Gray code.
- **Problem-solving:** Perform conversions between different binary codes.

Boolean Algebra

Boolean algebra is a mathematical system for analyzing and simplifying digital circuits. It deals with binary variables and logical operations.

- **Basic Postulates:**
 1. Closure: $A + B$ and $A \cdot B$ are in $\{0,1\}$.
 2. Commutative: $A + B = B + A$, $A \cdot B = B \cdot A$.
 3. Associative: $A + (B + C) = (A + B) + C$, $A \cdot (B \cdot C) = (A \cdot B) \cdot C$.

4. Distributive: $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$, $A + (B \cdot C) = (A + B) \cdot (A + C)$.

5. Identity: $A + 0 = A$, $A \cdot 1 = A$.

6. Complement: $A + \bar{A} = 1$, $A \cdot \bar{A} = 0$.

Theorems and Identities:

◦ Idempotence: $A + A = A$, $A \cdot A = A$.

◦ Absorption: $A + (A \cdot B) = A$, $A \cdot (A + B) = A$.

◦ Consensus: $(A \cdot B) + (\bar{A} \cdot C) + (B \cdot C) = (A \cdot B) + (\bar{A} \cdot C)$. (If $B \cdot C$ is redundant).

◦ De Morgan's Laws: $\overline{A + B} = \bar{A} \cdot \bar{B}$, $\overline{A \cdot B} = \bar{A} + \bar{B}$.

◦ Involution: $\bar{\bar{A}} = A$.

◦ Null Elements: $A + 1 = 1$, $A \cdot 0 = 0$.

• **Common Pitfalls:** Forgetting the dual of De Morgan's Law. Incorrectly applying consensus theorem.

• **Problem-solving:** Simplify Boolean expressions using theorems. Prove identities.

Booths Algorithm

Booth's algorithm is a multiplication algorithm that multiplies two signed binary numbers in 2's complement representation. It is efficient for numbers with long sequences of 0s or 1s.

• **Core Idea:** Instead of adding partial products for every '1' in the multiplier, it looks at pairs of bits in the multiplier.

◦ 00 → No operation (0 * multiplicand).

◦ 01 → Add multiplicand.

◦ 10 → Subtract multiplicand (add 2's complement).

◦ 11 → No operation (0 * multiplicand).

Steps:

1. Initialize Product (P) to 0, Multiplicand (M), Multiplier (Q), and an extra bit $Q_{-1} = 0$.

2. Repeat n times (where n is the number of bits in Q):

▪ Examine Q_0Q_{-1} :

▪ If 01: $P = P + M$.

▪ If 10: $P = P - M$ (add 2's complement of M).

▪ If 00 or 11: No operation.

▪ Arithmetic Right Shift (ASHR) the combined register PQQ_{-1} .

3. The final product is in PQ .

• **Common Pitfalls:** Incorrectly performing 2's complement subtraction. Errors in arithmetic right shift (especially for negative numbers). Forgetting the Q_{-1} bit.

• **Problem-solving:** Trace the algorithm step-by-step for given signed numbers.

Canonical Normal Form

Canonical normal forms are standard forms for Boolean expressions, ensuring uniqueness for a given function. They are useful for comparison and synthesis.

• **Canonical Sum of Products (SOP) / Minterm Canonical Form:** A sum of minterms. Each minterm is a product term where all variables appear exactly once, either in true or complemented form.

◦ A minterm is 1 for exactly one combination of inputs.

◦ Example: For $F(A, B) = A + B$, minterms are $\bar{A}B$, $A\bar{B}$, AB . So $F(A, B) = \bar{A}B + A\bar{B} + AB$.

◦ Notation: $\sum m(i, j, k, \dots)$, where i, j, k are decimal equivalents of minterms.

• **Canonical Product of Sums (POS) / Maxterm Canonical Form:** A product of maxterms. Each maxterm is a sum term where all variables appear exactly once, either in true or complemented form.

◦ A maxterm is 0 for exactly one combination of inputs.

◦ Example: For $F(A, B) = A \cdot B$, maxterms are $A + B$, $A + \bar{B}$, $\bar{A} + B$. So

$F(A, B) = (A + B) \cdot (A + \bar{B}) \cdot (\bar{A} + B)$.

◦ Notation: $\prod M(i, j, k, \dots)$, where i, j, k are decimal equivalents of maxterms.

• **Relationship:** If $F = \sum m(i, j, k)$, then $F = \sum m(\text{other indices})$. Also $F = \prod M(\text{other indices})$.

• **Common Pitfalls:** Confusing minterms with maxterms, and their corresponding sum/product forms. Incorrectly converting between SOP and POS canonical forms.

• **Problem-solving:** Convert truth tables to canonical SOP/POS. Convert between canonical SOP and POS.

Carry Generator

A carry generator is a combinational circuit used in adders, particularly in Carry Look-Ahead Adders, to quickly compute carry bits, thereby speeding up addition.

- **Core Idea:** It uses generate (G) and propagate (P) signals to calculate carries in parallel rather than waiting for them to ripple.
 - $G_i = A_i \cdot B_i$ (carry is generated if both input bits are 1).
 - $P_i = A_i \oplus B_i$ (carry is propagated if one input bit is 1).
 - $C_{i+1} = G_i + (P_i \cdot C_i)$ (carry out is generated or propagated from carry in).
- **Look-ahead Logic:** For a 4-bit block, the carry out C_4 can be expressed directly in terms of C_0 and the G_i, P_i terms as shown in the Adder section.
- **Problem-solving:** Derive carry equations for multi-bit carry look-ahead blocks.

Circuit Output

The circuit output refers to the final value(s) produced by a digital circuit based on its inputs and internal logic. It can be a single bit or a multi-bit word.

- **Combinational Circuits:** Output depends only on the present inputs.
- **Sequential Circuits:** Output depends on present inputs and past inputs (stored state).
- **Analysis:** For a given circuit diagram, determine the Boolean expression for the output(s) in terms of the inputs.
- **Problem-solving:** Derive truth tables or Boolean expressions from circuit diagrams. Evaluate output for specific input combinations.

Combinational Circuit

A combinational circuit is a type of digital circuit whose output depends solely on the current values of its inputs. It has no memory elements.

- **Characteristics:**
 - No feedback loops.
 - Output changes instantaneously with input changes (after propagation delay).
 - Examples: Adders, Subtractors, Decoders, Encoders, Multiplexers, Demultiplexers.
- **Design Steps:**
 1. Define the problem (inputs, outputs).
 2. Derive the truth table.
 3. Obtain simplified Boolean expressions (K-map, Quine-McCluskey).
 4. Draw the logic diagram.
- **Common Pitfalls:** Confusing combinational with sequential circuits. Overlooking propagation delays in timing analysis.
- **Problem-solving:** Design circuits from specifications. Analyze existing circuits.

Conjunctive Normal Form

Conjunctive Normal Form (CNF) is a standardized way of writing Boolean expressions as a conjunction (AND) of clauses, where each clause is a disjunction (OR) of literals.

- **Core Idea:** A product of sums (POS) form where each sum term (clause) contains one or more literals. It is not necessarily canonical (i.e., not all variables need to be present in each clause).
 - Example: $(A + \bar{B}) \cdot (\bar{A} + C)$.
- **Relationship to Canonical POS:** Canonical POS (minterm form) is a specific type of CNF where each clause is a maxterm (contains all variables).
- **Problem-solving:** Convert Boolean expressions to CNF.

Decoder

A decoder is a combinational circuit that converts n input lines into 2^n output lines. Only one output line is active (high or low) at any given time, corresponding to the binary value of the inputs.

- **Types:**
 - **Binary Decoder (n -to- 2^n):** E.g., 2-to-4 decoder, 3-to-8 decoder.
 - **BCD-to-7 Segment Decoder:** Converts BCD input to control a 7-segment display.
- **Enable Input:** Most decoders have an enable input (E). If E is inactive, all outputs are inactive.

- **Applications:** Memory address decoding, data demultiplexing, implementing Boolean functions.
 - An n -to- 2^n decoder can implement any n -variable Boolean function by ORing the appropriate minterms.
- **Common Pitfalls:** Incorrectly interpreting active-high vs. active-low outputs.
- **Problem-solving:** Design decoders. Use decoders to implement Boolean functions.

Digital Circuits

Digital circuits are electronic circuits that operate on discrete voltage levels, typically representing binary 0 and 1. They form the basis of all modern digital systems.

- **Categories:**
 - **Combinational Circuits:** Output depends only on current inputs (e.g., gates, adders, decoders).
 - **Sequential Circuits:** Output depends on current inputs and past inputs (memory elements) (e.g., flip-flops, counters, registers).
- **Logic Gates:** AND, OR, NOT, NAND, NOR, XOR, XNOR are the basic building blocks.
- **Key Properties:** Speed (propagation delay), Power Consumption, Fan-in/Fan-out, Noise Margin.
- **Problem-solving:** Analyze and design circuits using logic gates.

Digital Counter

A digital counter is a sequential circuit that cycles through a predefined sequence of states upon receiving input clock pulses. It's essentially a register that increments or decrements its stored value.

- **Types:**
 - **Asynchronous (Ripple) Counter:** Flip-flops are cascaded, and the output of one FF clocks the next. Simple to design but suffers from propagation delay.
 - **Synchronous Counter:** All flip-flops are clocked simultaneously by a common clock pulse. More complex design but faster and more reliable.
 - **Up/Down Counter:** Can count in increasing or decreasing order.
 - **Mod-N Counter:** Counts from 0 to $N-1$ and then resets.
- **Design:** Involves state diagrams, state tables, flip-flop excitation tables, and K-maps for logic minimization.
- **Formulas:** For an n -bit counter, it can count up to 2^n states (0 to $2^n - 1$).
- **Common Pitfalls:** Incorrectly determining the modulus of a counter. Errors in state table derivation or K-map minimization for synchronous counters.
- **Problem-solving:** Analyze existing counter circuits. Design synchronous counters for specific sequences.

Dual Function

The dual of a Boolean expression is obtained by interchanging OR and AND operations, and 0s and 1s. Variables and their complements remain unchanged.

- **Procedure:**
 1. Replace all '+' with '.' (OR with AND).
 2. Replace all '.' with '+' (AND with OR).
 3. Replace all '0' with '1'.
 4. Replace all '1' with '0'.
- **Example:** Dual of $F = A \cdot B + 0$ is $F^D = (A + B) \cdot 1$.
- **Principle of Duality:** If a Boolean identity is true, its dual is also true.
- **Common Pitfalls:** Accidentally complementing variables when finding the dual.
- **Problem-solving:** Find the dual of a given Boolean expression.

Finite State Machines (FSM)

Finite State Machines (FSMs) are mathematical models of computation used to design sequential circuits. They consist of a finite number of states, transitions between states, and actions based on inputs.

- **Components:** States, Inputs, Outputs, Transitions.
- **Types:**
 - **Mealy Machine:** Output depends on the present state AND the present input.
 - **Moore Machine:** Output depends only on the present state.
- **Representation:** State diagrams, state tables.
- **Design Steps:**

1. State Diagram.
 2. State Table.
 3. State Assignment (binary encoding).
 4. Flip-flop excitation table.
 5. K-maps for input equations and output equations.
 6. Logic diagram.
- **Common Pitfalls:** Confusing Mealy and Moore outputs. Errors in state assignment or deriving excitation equations.
 - **Problem-solving:** Design FSMs from specifications. Analyze given FSMs.

Fixed Point Representation

Fixed-point representation is a method of representing real numbers where the position of the binary point (radix point) is fixed. It's used for numbers with a limited range and high precision.

- **Format:** $I.F$ where I is the integer part and F is the fractional part.
 - For an n-bit number with k bits for the fractional part, the range is -2^{n-k-1} to $2^{n-k-1} - 2^{-k}$ (for signed 2's complement).
 - Value: $\sum_{i=-(k)}^{n-k-1} b_i 2^i$
- **Signed Fixed-Point:** Typically uses 2's complement for negative numbers.
- **Common Pitfalls:** Incorrectly determining the range or precision for a given bit allocation.
- **Problem-solving:** Convert decimal numbers to fixed-point binary and vice-versa. Perform arithmetic operations.

Flip Flop

A flip-flop is a basic 1-bit memory element (sequential circuit) that can store a binary value (0 or 1). It has two stable states and is often edge-triggered.

- **Types and Characteristics:**
 - **SR Flip-Flop:** Set-Reset.
 - Characteristic Equation: $Q_{next} = S + RQ$, $SR = 0$ (forbidden state if $SR=1$).
 - Excitation Table:

Q_n	Q_{n+1}	SR
0	0	0 X
0	1	1 0
1	0	0 1
1	1	X 0
 - **JK Flip-Flop:** J-K. Overcomes SR's forbidden state (toggle if $JK=11$).
 - Characteristic Equation: $Q_{next} = J\bar{Q} + \bar{K}Q$.
 - Excitation Table:

Q_n	Q_{n+1}	JK
0	0	0 X
0	1	1 X
1	0	X 1
1	1	X 0
 - **D Flip-Flop:** Data. Stores the input D when clocked.
 - Characteristic Equation: $Q_{next} = D$.
 - Excitation Table:

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1
 - **T Flip-Flop:** Toggle. Toggles its state if $T=1$, holds if $T=0$.
 - Characteristic Equation: $Q_{next} = T\bar{Q} + \bar{T}Q = T \oplus Q$.
 - Excitation Table:

Q_n	Q_{n+1}	T
0	0	0
0	1	1

1	0	1
1	1	0

- **Edge-Triggered vs. Level-Triggered:** Flip-flops are edge-triggered (positive or negative), latches are level-triggered.
- **Setup Time:** Data must be stable before clock edge.
- **Hold Time:** Data must be stable after clock edge.
- **Propagation Delay:** Time for output to change after clock edge.
- **Common Pitfalls:** Confusing characteristic equations with excitation tables. Misinterpreting edge vs. level triggering.
- **Problem-solving:** Analyze flip-flop behavior. Use flip-flops to design sequential circuits.

Floating Point Representation

Floating-point representation is a method to represent real numbers over a wider dynamic range than fixed-point, at the cost of precision. It uses a sign, exponent, and mantissa (significand).

- **Format:** $\pm M \times B^E$, where M is mantissa, B is base, E is exponent.
- **Normalization:** Mantissa is typically normalized to have a leading '1' (implicit in IEEE standard).
- **Bias:** Exponent is stored in biased form to allow representation of both positive and negative exponents without a separate sign bit. Biased exponent = Actual exponent + Bias.
- **Common Pitfalls:** Errors in converting between decimal and floating-point, especially with bias and normalization.
- **Problem-solving:** Convert numbers to/from floating-point format. Understand the implications of range and precision.

Functional Completeness

A set of logic gates is functionally complete if any Boolean function can be implemented using only gates from that set.

- **Minimum Functionally Complete Sets:**
 - {AND, OR, NOT}
 - {NAND} (NAND is universal)
 - {NOR} (NOR is universal)
 - {AND, NOT}
 - {OR, NOT}
- **Properties:** To be functionally complete, a set of gates must be able to implement NOT, AND, and OR (or NAND/NOR). This requires the ability to implement:
 - Inversion (NOT)
 - AND (or OR)
 - The set must not be monotonic (cannot implement NOT).
 - The set must not be self-dual (cannot implement NOT).
 - The set must not be linear (cannot implement AND/OR).
 - The set must not preserve 0 (cannot implement NOT).
 - The set must not preserve 1 (cannot implement NOT).
- **Common Pitfalls:** Incorrectly identifying functionally complete sets.
- **Problem-solving:** Determine if a given set of gates is functionally complete. Implement basic gates using a universal gate.

IEEE Representation

The IEEE 754 standard defines formats for representing floating-point numbers in computers, ensuring consistency across different platforms.

- **Single Precision (32-bit):**
 - 1 bit for Sign (S)
 - 8 bits for Exponent (E) - biased by 127
 - 23 bits for Mantissa (M) - implicit leading '1'
 - Value: $(-1)^S \times (1.M)_{base2} \times 2^{(E-127)}$
 - Range: Approx. $\pm 1.17 \times 10^{-38}$ to $\pm 3.4 \times 10^{38}$
- **Double Precision (64-bit):**
 - 1 bit for Sign (S)
 - 11 bits for Exponent (E) - biased by 1023

- 52 bits for Mantissa (M) - implicit leading '1'
- Value: $(-1)^S \times (1.M)_{base2} \times 2^{(E-1023)}$
- Special Values:**
 - Zero: $E = 0, M = 0$
 - Denormalized Numbers: $E = 0, M \neq 0$ (for very small numbers, implicit leading '0')
 - Infinity: $E = \text{all } 1\text{s}, M = 0$
 - NaN (Not a Number): $E = \text{all } 1\text{s}, M \neq 0$
- Common Pitfalls:** Forgetting the implicit leading '1' in the mantissa. Incorrectly applying the exponent bias. Handling special values.
- Problem-solving:** Convert decimal numbers to IEEE 754 format and vice-versa. Understand the representation of special values.

K Map (Karnaugh Map)

A K-map is a graphical method for simplifying Boolean expressions. It provides a systematic way to find the minimal sum-of-products (SOP) or product-of-sums (POS) form.

- Core Idea:** Arranges minterms (or maxterms) in a grid such that adjacent cells differ by only one bit, allowing visual identification of adjacent terms for grouping.
- Grouping Rules:**
 - Groups must be powers of 2 (1, 2, 4, 8, ...).
 - Groups must be rectangular or square.
 - Groups can wrap around the edges.
 - Groups should be as large as possible.
 - Every '1' (for SOP) or '0' (for POS) must be covered at least once.
- Prime Implicant (PI):** A product term obtained by combining the maximum possible number of adjacent cells in a K-map.
- Essential Prime Implicant (EPI):** A prime implicant that covers at least one minterm (or maxterm) that no other prime implicant covers. EPIs must be included in the minimal expression.
- Don't Cares (X):** Input combinations that never occur or whose output doesn't matter. They can be grouped with 1s (for SOP) or 0s (for POS) to make larger groups.
- Common Pitfalls:** Incorrect grouping (non-power of 2, non-rectangular). Not identifying all EPIs. Not finding the minimal set of PIs.
- Problem-solving:** Simplify Boolean expressions using K-maps for up to 4-5 variables.

Little Endian Big Endian

These terms refer to the byte ordering conventions used to store multi-byte data (like integers or floating-point numbers) in computer memory.

- Big-Endian:** The most significant byte (MSB) of a multi-byte data unit is stored at the lowest memory address.
 - Example: For $(0x12345678)_{16}$, at address 1000:

Address	Value
1000	12
1001	34
1002	56
1003	78
- Little-Endian:** The least significant byte (LSB) of a multi-byte data unit is stored at the lowest memory address.
 - Example: For $(0x12345678)_{16}$, at address 1000:

Address	Value
1000	78
1001	56
1002	34
1003	12
- Common Pitfalls:** Confusing the two, especially when dealing with memory dumps or network protocols.
- Problem-solving:** Determine how a given multi-byte value would be stored in memory under both endianness conventions.

Memory Interfacing

Memory interfacing involves connecting memory chips to a processor or other digital system, ensuring proper address decoding, data transfer, and control signaling.

- **Address Decoding:** Logic circuitry that selects the correct memory chip (or block within a chip) based on the address provided by the processor.
 - **Full Decoding:** Every unique address maps to a unique memory location.
 - **Partial Decoding:** Multiple addresses map to the same memory location (simpler but less efficient).
- **Chip Select (CS):** An input pin on memory chips that, when active, enables the chip for read/write operations. Address decoders generate the CS signal.
- **Output Enable (OE) / Read (RD):** Enables data output from memory.
- **Write Enable (WE) / Write (WR):** Enables data input to memory.
- **Memory Map:** A diagram showing how memory addresses are allocated to different memory devices.
- **Formulas:** For a memory chip with N address lines and M data lines, its capacity is $2^N \times M$ bits.
- **Common Pitfalls:** Incorrectly designing address decoding logic. Miscalculating memory capacity or address range.
- **Problem-solving:** Design address decoding circuits for given memory chips and address ranges. Calculate memory capacity.

Min No Gates

Minimizing the number of gates refers to finding the simplest possible logic circuit to implement a given Boolean function, often after minimizing the Boolean expression.

- **Core Idea:** Directly related to Boolean expression simplification (K-maps, Quine-McCluskey). Fewer terms and fewer literals generally mean fewer gates.
- **Considerations:**
 - **Gate Type:** NAND/NOR implementations often require fewer gates than AND/OR/NOT for universal logic.
 - **Fan-in:** Gates have a maximum number of inputs.
 - **Gate Cost:** Sometimes gates have different costs (e.g., XOR is more complex than AND).
- **Problem-solving:** After K-map simplification, draw the circuit using the minimal SOP/POS. Then convert to universal gates (NAND/NOR) if specified, and count the gates.

Min Products of Sum Form (MPOS)

The minimal Product of Sums (POS) form is the simplest Boolean expression representing a function as a product of sum terms (maxterms or sum literals).

- **Derivation:**
 1. Identify the '0's in the truth table or K-map.
 2. Group the '0's in the K-map (following K-map rules).
 3. Each group corresponds to a sum term.
 4. The product of these sum terms is the minimal POS expression.
- **Properties:** Each sum term is an implicant of the complement of the function.
- **Common Pitfalls:** Incorrectly forming sum terms from K-map groups (e.g., $\bar{A}$ for $A=1$).
- **Problem-solving:** Use K-maps to find the minimal POS expression.

Min Sum of Products Form (MSOP)

The minimal Sum of Products (SOP) form is the simplest Boolean expression representing a function as a sum of product terms (minterms or product literals).

- **Derivation:**
 1. Identify the '1's in the truth table or K-map.
 2. Group the '1's in the K-map (following K-map rules).
 3. Each group corresponds to a product term.
 4. The sum of these product terms is the minimal SOP expression.
- **Properties:** Each product term is an implicant of the function.
- **Common Pitfalls:** Not identifying all essential prime implicants. Not selecting the minimal set of prime implicants to cover all 1s.
- **Problem-solving:** Use K-maps to find the minimal SOP expression.

Multiplexer (MUX)

A multiplexer (MUX) is a combinational circuit that selects one of several input data lines and routes it to a single output line. The selection is controlled by a set of select lines.

- **Core Idea:** An n-to-1 MUX has n data inputs, $\log_2 n$ select lines, and 1 output.
- **Functionality:** If there are 's' select lines, it can select from 2^s data inputs.
- **Applications:** Data selection, parallel-to-serial conversion, implementing Boolean functions.
 - An n-variable Boolean function can be implemented using a $(n - 1)$ -variable MUX and external logic on the data inputs.
- **Formulas:** For a 2-to-1 MUX with select S, inputs I_0, I_1 : $Output = \bar{S}I_0 + SI_1$.
- **Common Pitfalls:** Incorrectly mapping inputs to select lines when implementing functions.
- **Problem-solving:** Design MUX circuits. Use MUXes to implement Boolean functions.

Number Representation

Number representation refers to the various ways in which numerical values (integers, real numbers) are stored and manipulated in digital systems.

- **Unsigned Binary:** All bits represent magnitude. Range for n bits: 0 to $2^n - 1$.
- **Signed Magnitude:** MSB is sign bit (0 for positive, 1 for negative), remaining bits are magnitude.
 - Range for n bits: $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$.
 - Two representations for zero (+0, -0).
- **1's Complement:** Negative numbers are obtained by inverting all bits of the positive number.
 - Range for n bits: $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$.
 - Two representations for zero (+0, -0).
 - Addition: Add normally, then add end-around carry to the sum.
- **2's Complement:** Negative numbers are obtained by inverting all bits and adding 1 to the result. Most common representation for signed integers.
 - Range for n bits: -2^{n-1} to $+(2^{n-1} - 1)$.
 - Unique representation for zero.
 - Addition/Subtraction: Perform binary addition directly. Overflow detection: if signs of inputs are same but sign of result is different.
- **Fixed-Point and Floating-Point:** Covered in their respective sections.
- **Common Pitfalls:** Errors in converting between signed representations. Incorrectly detecting overflow.
- **Problem-solving:** Convert numbers between different representations. Perform arithmetic operations and detect overflow.

Number System

A number system is a set of symbols and rules for representing numerical quantities. Digital systems primarily use binary, but conversions to/from decimal, octal, and hexadecimal are common.

- **Decimal (Base 10):** Digits 0-9.
- **Binary (Base 2):** Digits 0, 1.
- **Octal (Base 8):** Digits 0-7. Each octal digit corresponds to 3 binary bits.
- **Hexadecimal (Base 16):** Digits 0-9, A-F. Each hex digit corresponds to 4 binary bits.
- **Conversions:**
 - Decimal to Binary/Octal/Hex: Repeated division by base, collecting remainders.
 - Binary/Octal/Hex to Decimal: Sum of (digit * base^{position}).
 - Binary to Octal/Hex: Group bits (3 for octal, 4 for hex) and convert each group.
 - Octal/Hex to Binary: Convert each digit to its 3/4-bit binary equivalent.
- **Common Pitfalls:** Errors in base conversions, especially with fractional parts.
- **Problem-solving:** Perform conversions between any two number systems.

Prime Implicants

In Boolean algebra simplification (e.g. using K-maps), a prime implicant (PI) is a product term that cannot be combined with any other term to eliminate a literal.

- **Core Idea:** It represents a maximal group of adjacent 1s (or 0s for POS) in a K-map.
- **Essential Prime Implicant (EPI):** A prime implicant that covers at least one minterm (or maxterm) that no other prime implicant covers. All EPIs must be included in the minimal expression.

- **Redundant Prime Implicant:** A prime implicant whose covered minterms are all covered by other prime implicants (including EPIs).
- **Problem-solving:** Identify all PIs and EPIs from a K-map. Use them to derive minimal SOP/POS expressions.

ROM (Read-Only Memory)

ROM is a non-volatile memory device whose contents are programmed during manufacturing or by the user and cannot be easily changed during operation. It's used to store fixed programs or lookup tables.

- **Types:**
 - ◻ **Mask ROM:** Programmed during manufacturing.
 - ◻ **PROM (Programmable ROM):** User-programmable once.
 - ◻ **EPROM (Erasable PROM):** Erasable by UV light, reprogrammable.
 - ◻ **EEPROM (Electrically Erasable PROM):** Electrically erasable, reprogrammable.
 - ◻ **Flash Memory:** A type of EEPROM, block-erasable.
- **Structure:** Consists of an address decoder and an array of memory cells (fuses/transistors).
 - ◻ An n-input, m-output ROM can implement m different n-variable Boolean functions.
- **Applications:** Storing BIOS, lookup tables, implementing combinational logic functions.
- **Problem-solving:** Determine the size of ROM needed for a given function. Implement Boolean functions using ROM.

Reduction

Reduction in digital logic refers to the process of simplifying Boolean expressions or minimizing the number of gates/components in a circuit while maintaining its functionality.

- **Methods:**
 - ◻ **Boolean Algebra:** Applying theorems and postulates.
 - ◻ **Karnaugh Maps (K-maps):** Graphical method for up to 5-6 variables.
 - ◻ **Quine-McCluskey Algorithm:** Tabular method for more variables, suitable for computer implementation.
- **Goals:** Reduce cost (fewer gates), reduce power consumption, increase speed (fewer gate delays).
- **Common Pitfalls:** Not finding the absolute minimal form. Errors in applying simplification rules.
- **Problem-solving:** Apply various reduction techniques to simplify Boolean expressions.

Ripple Counter Operation

A ripple counter (asynchronous counter) is a sequential circuit where the output of one flip-flop serves as the clock input for the next flip-flop in the chain.

- **Operation:**
 - ◻ The first flip-flop (LSB) is clocked by the external clock.
 - ◻ Its output (Q) then clocks the next flip-flop, and so on.
 - ◻ This creates a "ripple" effect as the clock signal propagates through the stages.
- **Characteristics:**
 - ◻ **Simple Design:** Easy to construct.
 - ◻ **Propagation Delay:** The total delay accumulates through each flip-flop, leading to a maximum operating frequency limitation. $T_{total} = n \times T_{FF\ delay}$
 - ◻ **Glitches:** Due to varying propagation delays, intermediate states may appear briefly, causing glitches.
- **Modulus:** An n-bit ripple counter counts 2^n states (Mod- 2^n). To achieve Mod-N, external logic is needed to reset the counter at N.
- **Common Pitfalls:** Overlooking the cumulative propagation delay. Incorrectly determining the count sequence or modulus.
- **Problem-solving:** Analyze the timing and counting sequence of ripple counters. Calculate maximum operating frequency.

Shift Registers

A shift register is a sequential circuit capable of storing binary data and shifting it left or right by one or more bit positions on each clock pulse.

- **Types:**
 - ◻ **SISO (Serial-In, Serial-Out):** Data enters and exits serially.

- **SIPO (Serial-In, Parallel-Out):** Data enters serially, but all bits are available simultaneously at parallel outputs.
- **PISO (Parallel-In, Serial-Out):** Data enters in parallel, then shifts out serially.
- **PIPO (Parallel-In, Parallel-Out):** Data enters and exits in parallel.
- **Bidirectional Shift Register:** Can shift data both left and right.
- **Universal Shift Register:** Can perform all the above operations (SISO, SIPO, PISO, PIPO, load, hold).
- **Applications:** Data conversion (serial-to-parallel, parallel-to-serial), data manipulation, sequence generation, arithmetic operations (multiplication/division by powers of 2).
- **Structure:** Typically built using D flip-flops.
- **Common Pitfalls:** Incorrectly tracing data movement for different shift register types.
- **Problem-solving:** Analyze the operation of various shift registers. Design circuits using shift registers.

Static Hazard

A static hazard is a momentary undesirable glitch (a brief 0 or 1 pulse) in the output of a combinational circuit when the input changes, even though the output should remain constant.

- **Types:**
 - **Static-1 Hazard:** Output should remain 1 but momentarily goes to 0. Occurs when a K-map has two adjacent 1s that are covered by different prime implicants, and there's no overlapping prime implicant to cover the transition path.
 - **Static-0 Hazard:** Output should remain 0 but momentarily goes to 1. Occurs in POS implementations.
- **Cause:** Unequal propagation delays through different paths in the circuit.
- **Detection:** In a K-map for SOP, a static-1 hazard exists between two adjacent 1s if they are not covered by a common prime implicant.
- **Elimination:** Add redundant prime implicants to cover all adjacent pairs of 1s (for static-1 hazards) or 0s (for static-0 hazards). This means adding an extra AND gate (for SOP) or OR gate (for POS).
- **Common Pitfalls:** Forgetting to check for hazards in minimal circuits. Incorrectly adding redundant terms.
- **Problem-solving:** Identify static hazards in a given circuit or K-map. Eliminate hazards by adding redundant terms.

Synchronous Asynchronous Circuits

This distinction refers to how sequential circuits handle timing and state changes.

- **Synchronous Circuits:**
 - All memory elements (flip-flops) are controlled by a single, common clock signal.
 - State changes occur only at discrete clock edges.
 - Easier to design and debug due to predictable timing.
 - Less prone to race conditions and hazards.
 - Examples: Synchronous counters, shift registers, FSMs.
- **Asynchronous Circuits:**
 - Memory elements are not controlled by a common clock.
 - State changes occur in response to changes in input signals, without a clock.
 - Can be faster (no clock propagation delay) and consume less power.
 - Much harder to design and verify due to potential for race conditions and hazards.
 - Examples: Asynchronous (ripple) counters, latches, arbiters.
- **Common Pitfalls:** Confusing the characteristics and trade-offs of each type. Incorrectly identifying a circuit as synchronous or asynchronous.
- **Problem-solving:** Understand the timing behavior of both types of circuits.

Quick Formula Reference

- **Half Adder:**
 - $S = A \oplus B$
 - $C_{out} = A \cdot B$
- **Full Adder:**
 - $S = A \oplus B \oplus C_{in}$
 - $C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$
 - $C_{out} = (A \cdot B) + (B \cdot C_{in}) + (C_{in} \cdot A)$
- **Carry Look-Ahead Adder (stage i):**
 - $P_i = A_i \oplus B_i$
 - $G_i = A_i \cdot B_i$

- $S_i = P_i \oplus C_i$
- $C_{i+1} = G_i + (P_i \cdot C_i)$
- **Binary to Gray Code:** $G_i = B_i \oplus B_{i+1}$ (with $B_n = 0$ for MSB, or $G_{MSB} = B_{MSB}$, $G_i = B_i \oplus B_{i+1}$ for $i < MSB$).
- **Gray to Binary Code:** $B_i = G_i \oplus B_{i+1}$ (with $B_{MSB} = G_{MSB}$).
- **De Morgan's Laws:**
 - $A + B = \overline{\overline{A} \cdot \overline{B}}$
 - $A \cdot B = \overline{\overline{A} + \overline{B}}$
- **Absorption Law:**
 - $A + (A \cdot B) = A$
 - $A \cdot (A + B) = A$
- **Consensus Theorem:** $(A \cdot B) + (\overline{A} \cdot C) + (B \cdot C) = (A \cdot B) + (\overline{A} \cdot C)$
- **SR Flip-Flop Characteristic Equation:** $Q_{next} = S + \overline{R}Q$ (with $SR = 0$)
- **JK Flip-Flop Characteristic Equation:** $Q_{next} = J\overline{Q} + \overline{K}Q$
- **D Flip-Flop Characteristic Equation:** $Q_{next} = D$
- **T Flip-Flop Characteristic Equation:** $Q_{next} = T \oplus Q$
- **IEEE 754 Single Precision (32-bit):** $(-1)^S \times (1.M)_{base2} \times 2^{(E-127)}$
- **IEEE 754 Double Precision (64-bit):** $(-1)^S \times (1.M)_{base2} \times 2^{(E-1023)}$
- **Multiplexer (2-to-1):** $Output = \overline{S}I_0 + SI_1$
- **Ripple Counter Delay (n-bit):** $T_{total} = n \times T_{FF_delay}$
- **Memory Capacity (N address, M data):** $2^N \times M$ bits

Important Tips for GATE

1. **Master Boolean Algebra and K-Maps:** These are fundamental. Practice simplification extensively for both SOP and POS forms. Be quick and accurate in identifying prime implicants and essential prime implicants.
2. **Understand Number Representations Thoroughly:** Pay close attention to signed magnitude, 1's complement, and especially 2's complement. Practice conversions, arithmetic operations (addition, subtraction), and overflow detection for different bit lengths. IEEE 754 floating-point is a recurring topic; know the format, bias, and special values.
3. **Differentiate Combinational vs. Sequential Circuits:** Clearly understand their definitions, characteristics, and design methodologies. Be able to identify memory elements (flip-flops, latches) that make a circuit sequential.
4. **Memorize Flip-Flop Characteristic and Excitation Tables:** For SR, JK, D, and T flip-flops, these tables are crucial for analyzing and designing sequential circuits (counters, shift registers, FSMs). Don't confuse them.
5. **Practice FSM Design:** Be comfortable with state diagrams, state tables, state assignment, and deriving next-state and output equations using K-maps. Pay attention to Mealy vs. Moore machine characteristics.
6. **Beware of "Minimum Number of Gates" Questions:** These often require not just Boolean simplification but also considering universal gate implementations (NAND-only or NOR-only logic) and gate fan-in/fan-out constraints.
7. **Focus on Timing and Delays:** Understand propagation delays in combinational circuits (e.g., adders) and sequential circuits (e.g., ripple counters). Be able to calculate critical path delays and maximum operating frequencies.
8. **Don't Overlook Hazards:** Static hazards are a common trap. Learn how to detect them using K-maps and eliminate them by adding redundant terms.

4.1

Adder (9)

 **Practice Test:** Test 1 (12Q)

4.1.1 Adder: GATE CSE 1988 | Question: 4ii



Using binary full adders and other logic gates (if necessary), design an adder for adding 4-bit number (including sign) in 2's complement notation.

gate1988 digital-logic descriptive adder

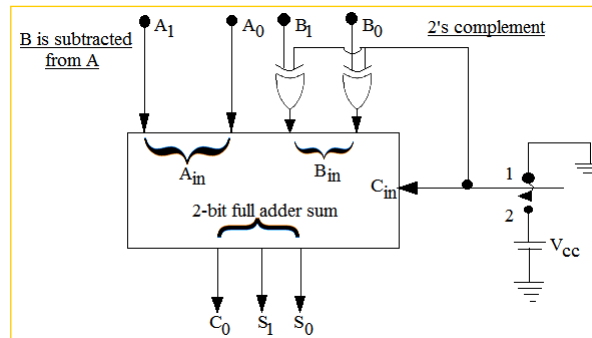
Answer key

4.1.2 Adder: GATE CSE 1990 | Question: 1-i



Fill in the blanks:

In the two bit full-adder/subtractor unit shown in below figure, when the switch is in position 2 _____ using _____ arithmetic.



gate1990 digital-logic adder fill-in-the-blanks

Answer key

4.1.3 Adder: GATE CSE 1997 | Question: 2.5

An N -bit carry lookahead adder, where N is a multiple of 4, employs ICs 74181 (4 bit ALU) and 74182 (4 bit carry lookahead generator).

The minimum addition time using the best architecture for this adder is

- A. proportional to N
- B. proportional to $\log N$
- C. a constant
- D. None of the above

gate1997 digital-logic normal adder

Answer key

4.1.4 Adder: GATE CSE 1999 | Question: 2.16

The number of full and half-adders required to add 16-bit numbers is

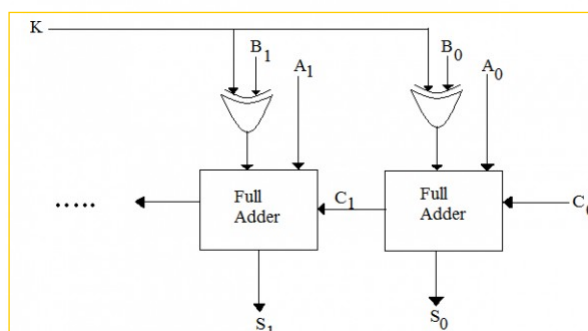
- A. 8 half-adders, 8 full-adders
- B. 1 half-adder, 15 full-adders
- C. 16 half-adders, 0 full-adders
- D. 4 half-adders, 12 full-adders

gate1999 digital-logic normal adder

Answer key

4.1.5 Adder: GATE CSE 2003 | Question: 46

Consider the ALU shown below.



If the operands are in 2's complement representation, which of the following operations can be performed by suitably setting the control lines K and C_0 only (+ and - denote addition and subtraction respectively)?

- A. $A + B$ and $A - B$, but not $A + 1$
- B. $A + B$ and $A + 1$, but not $A - B$
- C. $A + B$, but not $A - B$ or $A + 1$
- D. $A + B$, and $A - B$, and $A + 1$

gatecse-2003 digital-logic normal adder

Answer key

4.1.6 Adder: GATE CSE 2004 | Question: 62



A 4-bit carry look ahead adder, which adds two 4-bit numbers, is designed using AND, OR, NOT, NAND, NOR gates only. Assuming that all the inputs are available in both complemented and uncomplemented forms and the delay of each gate is one time unit, what is the overall propagation delay of the adder? Assume that the carry network has been implemented using two-level AND-OR logic.

- A. 4 time units
B. 6 time units
C. 10 time units
D. 12 time units

gatecse-2004 digital-logic normal adder

Answer key

4.1.7 Adder: GATE CSE 2015 | Set 2 | Question: 48



A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4-bit-ripple-carry binary adder is implemented by using four full adders. The total propagation time of this 4-bit binary adder in microseconds is _____.

gatecse-2015-set2 digital-logic adder normal numerical-answers

Answer key

4.1.8 Adder: GATE CSE 2016 | Set 1 | Question: 33



Consider a carry look ahead adder for adding two n -bit integers, built using gates of fan-in at most two. The time to perform addition using this adder is

- A. $\Theta(1)$
B. $\Theta(\log(n))$
C. $\Theta(\sqrt{n})$
D. $\Theta(n)$

gatecse-2016-set1 digital-logic adder normal

Answer key

4.1.9 Adder: GATE CSE 2016 | Set 2 | Question: 07



Consider an eight-bit ripple-carry adder for computing the sum of A and B where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____

gatecse-2016-set2 digital-logic adder normal numerical-answers

Answer key

4.2 Array Multiplier (2)

4.2.1 Array Multiplier: GATE CSE 1999 | Question: 1.21



The maximum gate delay for any output to appear in an array multiplier for multiplying two n bit numbers is

- A. $O(n^2)$
B. $O(n)$
C. $O(\log n)$
D. $O(1)$

gate1999 digital-logic normal array-multiplier

Answer key

4.2.2 Array Multiplier: GATE CSE 2003 | Question: 11



Consider an array multiplier for multiplying two n bit numbers. If each gate in the circuit has a unit delay, the total delay of the multiplier is

- A. $\Theta(1)$
B. $\Theta(\log n)$
C. $\Theta(n)$
D. $\Theta(n^2)$

gatecse-2003 digital-logic normal array-multiplier

Answer key

4.3.1 Binary Codes: GATE CSE 2006 | Question: 40



Consider numbers represented in 4-bit Gray code. Let $h_3 h_2 h_1 h_0$ be the Gray code representation of a number n and let $g_3 g_2 g_1 g_0$ be the Gray code of $(n + 1) \pmod{16}$ value of the number. Which one of the following functions is correct?

- A. $g_0(h_3 h_2 h_1 h_0) = \sum(1, 2, 3, 6, 10, 13, 14, 15)$
 B. $g_1(h_3 h_2 h_1 h_0) = \sum(4, 9, 10, 11, 12, 13, 14, 15)$
 C. $g_2(h_3 h_2 h_1 h_0) = \sum(2, 4, 5, 6, 7, 12, 13, 15)$
 D. $g_3(h_3 h_2 h_1 h_0) = \sum(0, 1, 6, 7, 10, 11, 12, 13)$

gatecse-2006 digital-logic number-representation binary-codes normal

Answer key

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (1Q) Weekly Quiz 1 (15Q) Weekly Quiz 2 (10Q)
 Weekly Quiz 3 (15Q) Weekly Quiz 4 (15Q)

4.4.1 Boolean Algebra: GATE CSE 1987 | Question: 1-II



The total number of Boolean functions which can be realised with four variables is:

- A. 4 B. 17 C. 256 D. 65,536

gate1987 digital-logic boolean-algebra functions combinatory

Answer key

4.4.2 Boolean Algebra: GATE CSE 1987 | Question: 12-a



The Boolean expression $A \oplus B \oplus A$ is equivalent to

- A. $AB + \bar{A} \bar{B}$ B. $\bar{A} B + A \bar{B}$
 C. B D. $\bar{A}$

gate1987 digital-logic boolean-algebra easy

Answer key

4.4.3 Boolean Algebra: GATE CSE 1988 | Question: 2-iii



Let $*$ be defined as a Boolean operation given as $x * y = \bar{x} \bar{y} + xy$ and let $C = A * B$. If $C = 1$ then prove that $A = B$.

gate1988 digital-logic descriptive boolean-algebra

Answer key

4.4.4 Boolean Algebra: GATE CSE 1989 | Question: 4-x



A switching function is said to be neutral if the number of input combinations for which its value is 1 is equal to the number of input combinations for which its value is 0. Compute the number of neutral switching functions of n variables (for a given n).

gate1989 descriptive digital-logic boolean-algebra

Answer key

4.4.5 Boolean Algebra: GATE CSE 1989 | Question: 5-a



Find values of Boolean variables A, B, C which satisfy the following equations:

- $A + B = 1$
- $AC = BC$

- $A + C = 1$
- $AB = 0$

gate1989 descriptive digital-logic boolean-algebra

Answer key

4.4.6 Boolean Algebra: GATE CSE 1992 | Question: 02-i

The operation which is commutative but not associative is:

- A. AND B. OR C. EX-OR D. NAND

gate1992 easy digital-logic boolean-algebra multiple-selects

Answer key

4.4.7 Boolean Algebra: GATE CSE 1994 | Question: 4

A. Let $*$ be a Boolean operation defined as $A * B = AB + \bar{A}\bar{B}$. If $C = A * B$ then evaluate and fill in the blanks:

- i. $A * A =$
ii. $C * A =$

B. Solve the following boolean equations for the values of A, B and C :

$$\begin{aligned} AB + \bar{A}C &= 1 \\ AC + B &= 0 \end{aligned}$$

gate1994 digital-logic normal boolean-algebra descriptive

Answer key

4.4.8 Boolean Algebra: GATE CSE 1995 | Question: 2.5

What values of A, B, C and D satisfy the following simultaneous Boolean equations?

$$\bar{A} + AB = 0, AB = AC, AB + A\bar{C} + CD = \bar{C}D$$

- A. $A = 1, B = 0, C = 0, D = 1$ B. $A = 1, B = 1, C = 0, D = 0$
C. $A = 1, B = 0, C = 1, D = 1$ D. $A = 1, B = 0, C = 0, D = 0$

gate1995 digital-logic boolean-algebra easy

Answer key

4.4.9 Boolean Algebra: GATE CSE 1997 | Question: 2-1

Let $*$ be defined as $x * y = \bar{x} + y$. Let $z = x * y$. Value of $z * x$ is

- A. $\bar{x} + y$ B. x C. 0 D. 1

gate1997 digital-logic normal boolean-algebra

Answer key

4.4.10 Boolean Algebra: GATE CSE 1998 | Question: 1.13

What happens when a bit-string is XORed with itself n -times as shown:

$$[B \oplus (B \oplus (B \oplus (B \dots n \text{ times})))]$$

- A. complements when n is even B. complements when n is odd
C. divides by 2^n always D. remains unchanged when n is even

gate1998 digital-logic normal boolean-algebra

Answer key

4.4.11 Boolean Algebra: GATE CSE 1998 | Question: 2.8



Which of the following operations is commutative but not associative?

- A. AND B. OR C. NAND D. EXOR

gate1998 digital-logic easy boolean-algebra

Answer key >

4.4.12 Boolean Algebra: GATE CSE 1999 | Question: 1.7



Which of the following expressions is not equivalent to $\bar{x}$?

- A. $x \text{ NAND } x$ B. $x \text{ NOR } x$ C. $x \text{ NAND } 1$ D. $x \text{ NOR } 1$

gate1999 digital-logic easy boolean-algebra

Answer key >

4.4.13 Boolean Algebra: GATE CSE 2000 | Question: 2.10



The simultaneous equations on the Boolean variables x, y, z and w ,

- $x + y + z = 1$
- $xy = 0$
- $xz + w = 1$
- $xy + \bar{z}\bar{w} = 0$

have the following solution for x, y, z and w , respectively:

- A. 0 1 0 0 B. 1 1 0 1
C. 1 0 1 1 D. 1 0 0 0

gatecse-2000 digital-logic boolean-algebra easy

Answer key >

4.4.14 Boolean Algebra: GATE CSE 2002 | Question: 2-3



Let $f(A, B) = A' + B$. Simplified expression for function $f(f(x + y, y), z)$ is

- A. $x' + z$ B. xyz C. $xy' + z$ D. None of the above

gatecse-2002 digital-logic boolean-algebra normal

Answer key >

4.4.15 Boolean Algebra: GATE CSE 2004 | Question: 17



A Boolean function $x'y' + xy + x'y$ is equivalent to

- A. $x' + y'$ B. $x + y$ C. $x + y'$ D. $x' + y$

gatecse-2004 digital-logic easy boolean-algebra

Answer key >

4.4.16 Boolean Algebra: GATE CSE 2007 | Question: 32



Let $f(w, x, y, z) = \sum (0, 4, 5, 7, 8, 9, 13, 15)$. Which of the following expressions are NOT equivalent to f ?

- P: $x'y'z' + w'xy' + wy'z + xz$
Q: $w'y'z' + wx'y' + xz$
R: $w'y'z' + wx'y' + xyz + xy'z$
S: $x'y'z' + wx'y' + w'y$

- A. P only B. Q and S C. R and S D. S only

gatecse-2007 digital-logic normal boolean-algebra

Answer key

4.4.17 Boolean Algebra: GATE CSE 2007 | Question: 33



Define the connective $*$ for the Boolean variables X and Y as:

$$X * Y = XY + X'Y'$$

Let $Z = X * Y$. Consider the following expressions P , Q and R :

$$\begin{aligned} P : X &= Y * Z, \\ Q : Y &= X * Z, \\ R : X * Y * Z &= 1 \end{aligned}$$

Which of the following is **TRUE**?

- A. Only P and Q are valid.
C. Only P and R are valid.

- B. Only Q and R are valid.
D. All P , Q , R are valid.

gatecse-2007 digital-logic normal boolean-algebra

Answer key

4.4.18 Boolean Algebra: GATE CSE 2008 | Question: 26



If P, Q, R are Boolean variables, then

$(P + \bar{Q})(P \cdot \bar{Q} + P \cdot R)(\bar{P} \cdot \bar{R} + \bar{Q})$ simplifies to

A. $P \cdot \bar{Q}$

B. $P \cdot \bar{R}$

C. $P \cdot \bar{Q} + R$

D. $P \cdot \bar{R} + Q$

gatecse-2008 easy digital-logic boolean-algebra

Answer key

4.4.19 Boolean Algebra: GATE CSE 2012 | Question: 6



The truth table	X	Y	(X,Y)
	0	0	0
	0	1	0
	1	0	1
	1	1	1

represents the Boolean function

A. X

B. $X + Y$

C. $X \oplus Y$

D. Y

gatecse-2012 digital-logic easy boolean-algebra

Answer key

4.4.20 Boolean Algebra: GATE CSE 2013 | Question: 21



Which one of the following expressions does **NOT** represent exclusive NOR of x and y ?

- A. $xy + x'y'$
C. $x' \oplus y$

- B. $x \oplus y'$
D. $x' \oplus y'$

gatecse-2013 digital-logic easy boolean-algebra

Answer key

4.4.21 Boolean Algebra: GATE CSE 2014 | Set 3 | Question: 55



Let $\oplus$ denote the exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q :

$$F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$$

The equivalent expression for F is

- A. $P + Q$
C. $P \oplus Q$

- B. $\overline{P + Q}$
D. $\overline{P \oplus Q}$

gatecse-2014-set3 digital-logic normal boolean-algebra

Answer key

4.4.22 Boolean Algebra: GATE CSE 2015 | Set 2 | Question: 37



The number of min-terms after minimizing the following Boolean expression is _____.

$$[D' + AB' + A'C + AC'D + A'C'D']'$$

gatecse-2015-set2 digital-logic boolean-algebra normal numerical-answers

Answer key

4.4.23 Boolean Algebra: GATE CSE 2016 | Set 1 | Question: 06



Consider the Boolean operator $\#$ with the following properties :

$x \# 0 = x, x \# 1 = \bar{x}, x \# x = 0$ and $x \# \bar{x} = 1$. Then $x \# y$ is equivalent to

- A. $x\bar{y} + \bar{x}y$
C. $\bar{x}y + xy$

- B. $x\bar{y} + \bar{x}\bar{y}$
D. $xy + \bar{x}\bar{y}$

gatecse-2016-set1 digital-logic boolean-algebra easy

Answer key

4.4.24 Boolean Algebra: GATE CSE 2016 | Set 2 | Question: 08



Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator.

Which one of the following must always be TRUE?

- A. $x_1 x_2 x_3 x_4 = 0$
B. $x_1 x_3 + x_2 = 0$
C. $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$
D. $x_1 + x_2 + x_3 + x_4 = 0$

gatecse-2016-set2 digital-logic boolean-algebra normal

Answer key

4.4.25 Boolean Algebra: GATE CSE 2017 | Set 2 | Question: 27



If w, x, y, z are Boolean variables, then which one of the following is INCORRECT?

- A. $wx + w(x + y) + x(x + y) = x + wy$
C. $(w\bar{x}(y + xz) + \bar{w}\bar{x})y = x\bar{y}$

- B. $\overline{w\bar{x}(y + z)} + \bar{w}x = \bar{w} + x + \bar{y}z$
D. $(w + y)(wxy + wyz) = wxy + wyz$

gatecse-2017-set2 digital-logic boolean-algebra normal

Answer key

4.4.26 Boolean Algebra: GATE CSE 2018 | Question: 4



Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT?

- A. $\overline{P \oplus Q} = P \odot Q$
B. $\overline{P \oplus Q} = P \odot Q$
C. $\overline{P \oplus Q} = P \oplus Q$
D. $P \oplus \overline{P \oplus Q} = (P \odot \overline{P \odot Q})$

gatecse-2018 digital-logic normal boolean-algebra one-mark

Answer key

4.4.27 Boolean Algebra: GATE CSE 2019 | Question: 6



Which one of the following is NOT a valid identity?

A. $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

B. $(x + y) \oplus z = x \oplus (y + z)$

C. $x \oplus y = x + y$, if $xy = 0$

D. $x \oplus y = (xy + x'y')$

gatecse-2019 digital-logic boolean-algebra one-mark

Answer key

4.4.28 Boolean Algebra: GATE CSE 2021 | Set 1 | Question: 42



Consider the following Boolean expression.

$$F = (X + Y + Z)(\bar{X} + Y)(\bar{Y} + Z)$$

Which of the following Boolean expressions is/are equivalent to $\bar{F}$ (complement of F)?

A. $(\bar{X} + \bar{Y} + \bar{Z})(X + \bar{Y})(Y + \bar{Z})$

B. $X\bar{Y} + \bar{Z}$

C. $(X + \bar{Z})(\bar{Y} + \bar{Z})$

D. $X\bar{Y} + Y\bar{Z} + \bar{X}\bar{Y}\bar{Z}$

gatecse-2021-set1 multiple-selects digital-logic boolean-algebra two-marks

Answer key

4.4.29 Boolean Algebra: GATE CSE 2024 | Set 2 | Question: 20



For a Boolean variable x , which of the following statements is/are FALSE?

A. $x.1 = x$

B. $x + 1 = x$

C. $x \cdot x = 0$

D. $x + \bar{x} = 1$

gatecse-2024-set2 digital-logic boolean-algebra easy multiple-selects one-mark

Answer key

4.4.30 Boolean Algebra: GATE CSE 2025 | Set 1 | Question: 14



Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables?

A. $X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)$

B. $X(a, b, X(a, b, c)) = X(a, b, c)$

C. $X(a, b, X(a, c, d)) = (X(a, b, a) \text{ AND } X(c, d, c))$

D. $X(a, b, c) = X(a, X(a, b, c), X(a, c, c))$

gatecse2025-set1 digital-logic boolean-algebra multiple-selects one-mark

Answer key

4.4.31 Boolean Algebra: GATE CSE 2025 | Set 2 | Question: 40



Which of the following Boolean algebraic equation(s) is/are CORRECT?

A. $\bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC = BC + \bar{B}\bar{C} + \bar{A}B$

B. $AB + \bar{A}C + BC = AB + \bar{A}C$

C. $(A + C)(\bar{A} + B) = AB + \bar{A}C$

D. $(A + \bar{B} + \bar{D})(C + D)(\bar{A} + C + D)(A + B + \bar{D}) = \bar{A}D + \bar{C}\bar{D}$

gatecse2025-set2 digital-logic boolean-algebra multiple-selects easy two-marks

Answer key

4.4.32 Boolean Algebra: GATE CSE 2026 | Set 2 | Question: 6



Which one of the following options is **not** a property of Boolean Algebra?

Note: $+$ is OR operation, $\cdot$ is AND operation, and $'$ is NOT operation

A. $a + b = b + a$

B. $a \cdot a' = 1$

C. $a + a' = 1$

D. $a \cdot b = b \cdot a$

gatecse-2026-set2 digital-logic boolean-algebra easy one-mark

Answer key

4.4.33 Boolean Algebra: GATE IT 2004 | Question: 44



The function $\bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C$ is equivalent to

A. $A\bar{C} + AB + \bar{A}\bar{C}$

B. $A\bar{B} + A\bar{C} + \bar{A}\bar{C}$

C. $\bar{A}B + A\bar{C} + \bar{A}\bar{B}$

D. $\bar{A}B + AC + A\bar{B}$

gateit-2004 digital-logic boolean-algebra easy

Answer key

4.4.34 Boolean Algebra: GATE IT 2005 | Question: 7



Which of the following expressions is equivalent to $(A \oplus B) \oplus C$

A. $(A + B + C)(\bar{A} + \bar{B} + \bar{C})$

B. $(A + B + C)(\bar{A} + \bar{B} + C)$

C. $ABC + \bar{A}(B \oplus C) + \bar{B}(A \oplus C)$

D. None of these

gateit-2005 digital-logic normal boolean-algebra

Answer key

4.5

Booths Algorithm (7)

Practice Test: Test (6Q)

4.5.1 Booths Algorithm: GATE CSE 1990 | Question: 8b



State the Booth's algorithm for multiplication of two numbers. Draw a block diagram for the implementation of the Booth's algorithm for determining the product of two 8-bit signed numbers.

gate1990 descriptive digital-logic booths-algorithm

Answer key

4.5.2 Booths Algorithm: GATE CSE 1996 | Question: 1.23



Booth's algorithm for integer multiplication gives worst performance when the multiplier pattern is

A. 101010...1010

B. 100000...0001

C. 111111...1111

D. 011111...1110

gate1996 digital-logic booths-algorithm normal

Answer key

4.5.3 Booths Algorithm: GATE CSE 1999 | Question: 1.20



Booth's coding in 8 bits for the decimal number -57 is:

A. $0 - 100 + 1000$

B. $0 - 100 + 100 - 1$

C. $0 - 1 + 100 - 10 + 1$

D. $00 - 10 + 100 - 1$

gate1999 digital-logic number-representation booths-algorithm normal

Answer key

4.5.4 Booths Algorithm: GATE CSE 2025 | Set 2 | Question: 22



The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm:

M: 1100110111101101 and Q: 1010010010101010

The total number of addition and subtraction operations to be performed is _____. (Answer in integer)

gatecse2025-set2 digital-logic booths-algorithm numerical-answers one-mark

Answer key

4.5.5 Booths Algorithm: GATE IT 2005 | Question: 8



Using Booth's Algorithm for multiplication, the multiplier -57 will be recoded as

A. 0 -1 00 1 00 -1

B. 1 1 0 0 0 1 1 1

C. 0 -1 0 0 1 0 0 0

D. 0 1 0 0 -1 0 0 1

gateit-2005 digital-logic booths-algorithm normal

Answer key

4.5.6 Booths Algorithm: GATE IT 2006 | Question: 38



When multiplicand Y is multiplied by multiplier $X = x_{n-1}x_{n-2} \dots x_0$ using bit-pair recoding in Booth's algorithm, partial products are generated according to the following table.

Row	x_{i+1}	x_i	x_{i-1}	Partial Product
1	0	0	0	0
2	0	0	1	Y
3	0	1	0	Y
4	0	1	1	2Y
5	1	0	0	?
6	1	0	1	-Y
7	1	1	0	-Y
8	1	1	1	?

The partial products for rows 5 and 8 are

A. $2Y$ and Y

B. $-2Y$ and $2Y$

C. $-2Y$ and 0

D. 0 and Y

gateit-2006 digital-logic booths-algorithm difficult

Answer key

4.5.7 Booths Algorithm: GATE IT 2008 | Question: 42



The two numbers given below are multiplied using the Booth's algorithm.

Multiplicand : 0101 1010 1110 1110

Multiplier: 0111 0111 1011 1101

How many additions/Subtractions are required for the multiplication of the above two numbers?

A. 6

B. 8

C. 10

D. 12

gateit-2008 digital-logic booths-algorithm normal

Answer key

4.6

Canonical Normal Form (10)

Practice Test: Test 1 (11Q)

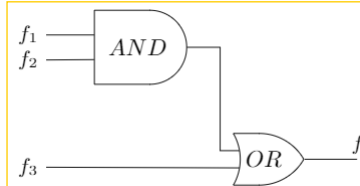
4.6.1 Canonical Normal Form: GATE CSE 1988 | Question: 2-v



Three switching functions f_1 , f_2 and f_3 are expressed below as sum of minterms.

- $f_1(w, x, y, z) = \sum 0, 1, 2, 3, 5, 12$
- $f_2(w, x, y, z) = \sum 0, 1, 2, 10, 13, 14, 15$
- $f_3(w, x, y, z) = \sum 2, 4, 5, 8$

Express the function f realised by the circuit shown in the below figure as the sum of minterms (in decimal notation).



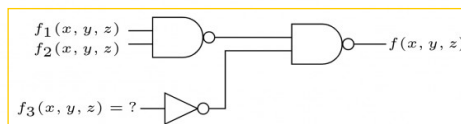
gate1988 descriptive digital-logic easy circuit-output canonical-normal-form digital-circuits

Answer key

4.6.2 Canonical Normal Form: GATE CSE 2002 | Question: 2-1



Consider the following logic circuit whose inputs are functions f_1, f_2, f_3 and output is f



Given that

- $f_1(x, y, z) = \sum(0, 1, 3, 5)$
- $f_2(x, y, z) = \sum(6, 7)$, and
- $f(x, y, z) = \sum(1, 4, 5)$.

f_3 is

- A. $\sum(1, 4, 5)$
- B. $\sum(6, 7)$
- C. $\sum(0, 1, 3, 5)$
- D. None of the above

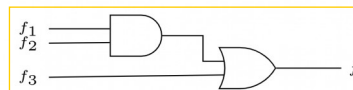
gatecse-2002 digital-logic normal canonical-normal-form circuit-output

Answer key

4.6.3 Canonical Normal Form: GATE CSE 2008 | Question: 8



Given f_1, f_3 and f in canonical sum of products form (in decimal) for the circuit



$$\begin{aligned} f_1 &= \sum m(4, 5, 6, 7, 8) \\ f_3 &= \sum m(1, 6, 15) \\ f &= \sum m(1, 6, 8, 15) \end{aligned}$$

then f_2 is

- A. $\sum m(4, 6)$
- B. $\sum m(4, 8)$
- C. $\sum m(6, 8)$
- D. $\sum m(4, 6, 8)$

gatecse-2008 digital-logic canonical-normal-form easy

Answer key

4.6.4 Canonical Normal Form: GATE CSE 2010 | Question: 6



The minterm expansion of $f(P, Q, R) = PQ + QR + PR$ is

- A. $m_2 + m_4 + m_6 + m_7$ B. $m_0 + m_1 + m_3 + m_5$
 C. $m_0 + m_1 + m_6 + m_7$ D. $m_2 + m_3 + m_4 + m_5$

gatecse-2010 digital-logic canonical-normal-form normal

Answer key

4.6.5 Canonical Normal Form: GATE CSE 2015 | Set 3 | Question: 43



The total number of prime implicants of the function $f(w, x, y, z) = \sum(0, 2, 4, 5, 6, 10)$ is _____

gatecse-2015-set3 digital-logic canonical-normal-form normal numerical-answers

Answer key

4.6.6 Canonical Normal Form: GATE CSE 2015 | Set 3 | Question: 44



Given the function $F = P' + QR$ where F is a function in three Boolean variables P, Q and R and $P' = \neg P$, consider the following statements.

- (S1) $F = \sum(4, 5, 6)$
 (S2) $F = \sum(0, 1, 2, 3, 7)$
 (S3) $F = \Pi(4, 5, 6)$
 (S4) $F = \Pi(0, 1, 2, 3, 7)$

Which of the following is true?

- A. (S1)-False, (S2)-True, (S3)-True, (S4)-False
 B. (S1)-True, (S2)-False, (S3)-False, (S4)-True
 C. (S1)-False, (S2)-False, (S3)-True, (S4)-True
 D. (S1)-True, (S2)-True, (S3)-False, (S4)-False

gatecse-2015-set3 digital-logic canonical-normal-form normal

Answer key

4.6.7 Canonical Normal Form: GATE CSE 2019 | Question: 30



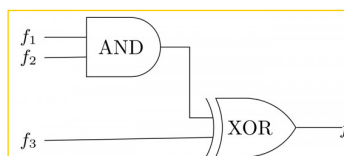
Consider three 4-variable functions f_1, f_2 and f_3 , which are expressed in sum-of-minterms as

$$f_1 = \sum(0, 2, 5, 8, 14)$$

$$f_2 = \sum(2, 3, 6, 8, 14, 15)$$

$$f_3 = \sum(2, 7, 11, 14)$$

For the following circuit with one AND gate and one XOR gate the output function f can be expressed as:



- A. $\sum(7, 8, 11)$ B. $\sum(2, 7, 8, 11, 14)$
 C. $\sum(2, 14)$ D. $\sum(0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$

gatecse-2019 digital-logic digital-circuits two-marks canonical-normal-form

Answer key

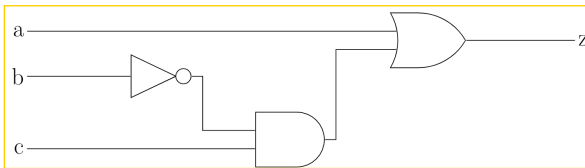
4.6.8 Canonical Normal Form: GATE CSE 2019 | Question: 50



What is the minimum number of 2-input NOR gates required to implement a 4-variable function expressed in sum-of-minterms form as $f = \sum(0, 2, 5, 7, 8, 10, 13, 15)$? Assume that all the inputs and their complements are available. Answer: _____

Answer key

4.6.9 Canonical Normal Form: GATE CSE 2020 | Question: 28

Consider the Boolean function $z(a, b, c)$.

Which one of the following minterm lists represents the circuit given above?

A. $z = \sum(0, 1, 3, 7)$
 C. $z = \sum(2, 4, 5, 6, 7)$

B. $z = \sum(1, 4, 5, 6, 7)$
 D. $z = \sum(2, 3, 5)$

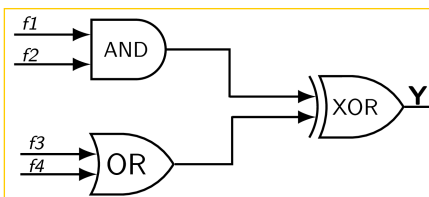
gatecse-2020 digital-logic canonical-normal-form two-marks

Answer key

4.6.10 Canonical Normal Form: GATE CSE 2024 | Set 2 | Question: 40

Consider 4-variable functions f_1, f_2, f_3, f_4 expressed in sum-of-minterms form as given below.

$$\begin{aligned} f_1 &= \sum(0, 2, 3, 5, 7, 8, 11, 13) \\ f_2 &= \sum(1, 3, 5, 7, 11, 13, 15) \\ f_3 &= \sum(0, 1, 4, 11) \\ f_4 &= \sum(0, 2, 6, 13) \end{aligned}$$



With respect to the circuit given above, which of the following options is/are CORRECT?

A. $Y = \sum(0, 1, 2, 11, 13)$
 C. $Y = \sum(0, 1, 2, 3, 4, 5, 6, 7)$

B. $Y = \Pi(3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 15)$
 D. $Y = \Pi(8, 9, 10, 11, 12, 13, 14, 15)$

gatecse-2024-set2 digital-logic canonical-normal-form multiple-selects two-marks

Answer key

4.7

Carry Generator (2)

4.7.1 Carry Generator: GATE CSE 2006 | Question: 36



Given two three bit numbers $a_2a_1a_0$ and $b_2b_1b_0$ and c the carry in, the function that represents the carry generate function when these two numbers are added is:

- A. $a_2b_2 + a_2a_1b_1 + a_2a_1a_0b_0 + a_2a_0b_1b_0 + a_1b_2b_1 + a_1a_0b_2b_0 + a_0b_2b_1b_0$
 B. $a_2b_2 + a_2b_1b_0 + a_2a_1b_1b_0 + a_1a_0b_2b_1 + a_1a_0b_2 + a_1a_0b_2b_0 + a_2a_0b_1b_0$
 C. $a_2 + b_2 + (a_2 \oplus b_2)(a_1 + b_1 + (a_1 \oplus b_1) + (a_0 + b_0))$
 D. $a_2b_2 + \overline{a_2}a_1b_1 + \overline{a_2}\overline{a_1}a_0b_0 + \overline{a_2}a_0\overline{b_1}b_0 + a_1\overline{b_2}b_1 + \overline{a_1}a_0\overline{b_2}b_0 + a_0\overline{b_2}b_1b_0$

gatecse-2006 digital-logic normal carry-generator adder

Answer key